

Ground then Navigate: Language-guided Navigation in Dynamic Scenes

Thesis submitted in partial fulfillment
of the requirements for the degree of

Master of Science
in
Computer Science and Engineering
by Research

by

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2019121011

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June, 2023

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CERTIFICATE

It is certified that the work contained in this thesis, titled “Ground then Navigate: Language-guided Navigation in Dynamic Scenes” by Varun Chhangani, has been carried out under my supervision and is not submitted elsewhere for a degree.

August 3, 2026

Adviser: Dr. Vineet Gandhi

उद्यमेन हि सिध्यन्ति कार्याणि न मनोरथैः।
न हि सुप्तस्य सिंहस्य प्रविशन्ति मुखे मृगाः॥

Acknowledgments

I want to thank my advisor, Dr. Vineet Gandhi, for his constant support and guidance throughout this research. His clarity of thought, insistence on interpretable methods, and patience through many iterations of ideas were pivotal in shaping this thesis. I am grateful for the chance to work under his mentorship and aim to make him proud as I move on to the next step of my career.

I would also like to thank Prof. K. Madhava Krishna and the faculty and researchers at the Kohli Center on Intelligent Systems (KCIS) and the Robotics Research Center (RRC) for creating an environment that made this work possible. The courses and discussions around vision, robotics, and autonomous systems were foundational to this thesis.

I am deeply grateful to my collaborators Kanishk Jain, Amogh Tiwari, and KV Aditya Srivatsa for their partnership on the CARLA-NAV dataset and the navigation pipeline. Collecting and verifying hundreds of episodes was a team effort, and this thesis would not exist without their contributions. I also thank everyone who helped with annotation, verification, and technical advice along the way.

Lastly, I want to thank my family for their constant love, moral support, and valuable life advice, which helped me cope with my studies and research.

Abstract

We investigate the Vision-and-Language Navigation (VLN) problem in the context of autonomous driving in outdoor settings. Existing VLN methods typically pose navigation as a node-selection problem on a discrete connected graph of the environment, or as grounding on a single static image. Both assumptions break down in dynamically changing outdoor scenes: a discrete map is often unavailable, and the region referred to by a language command may not yet be visible, may be occluded, or may leave the frame as the vehicle moves.

This thesis addresses these limitations by explicitly grounding the navigable regions corresponding to a textual command. At each timestamp, the model predicts a segmentation mask for the intermediate or final navigable region. The approach moves toward continuity in action space, provides interpretability through visual feedback, and supports finer maneuvers such as “park between the two cars”.

To enable training and live evaluation, we introduce CARLA-NAV, a novel meta-dataset built on the CARLA simulator. It comprises pre-recorded training sequences and a live environment for validation and testing, spanning multiple maps, weather conditions, and dynamic traffic. We further propose a multi-task network that jointly predicts navigable-region and short-term future-trajectory masks from multi-frame visual features, a historical contextual trajectory, and the language command, using a CLIP backbone. We also document earlier ViT and Conv3D (X3D) architectures evaluated with displacement metrics and stratified command ablations.

Experiments on CARLA-NAV (25 validation / 34 test episodes) show that combining multiple frames with historical trajectory context improves human-evaluated Task Completion, Fréchet Distance, and normalized Dynamic Time Warping over single-image and non-contextual baselines, with Wilson confidence intervals reported for Task Completion. We also provide a formal analysis of grounding-first control and an extended evaluation protocol (map-unseen splits, safety infractions, trajectory-head ablation, and runtime). Relative to discrete-action outdoor VLN, the contribution is continuous-control region grounding with live episode evaluation in dynamic simulation.

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List of Outstanding Items

- **TODO:** Add the train-split command list and extend Tables 3.2–3.4 with a Train column (and full-dataset totals). Held-out val/test counts above are complete. . . . 18

List of Acronyms

ADE	Average Displacement Error
ADE-Match	Average Displacement Error (coupling-based)
ASPP	Atrous Spatial Pyramid Pooling
BCE	Binary Cross-Entropy
BERT	Bidirectional Encoder Representations from Transformers
CARLA	Car Learning to Act
CIRL	Controllable Imitative Reinforcement Learning
CLIP	Contrastive Language–Image Pretraining
CNN	Convolutional Neural Network
DUET	Dual-scale Graph Transformer
EnvDrop	Environmental Dropout
FDE	Final Displacement Error
GloVe	Global Vectors for Word Representation
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IoU	Intersection over Union
LiDAR	Light Detection and Ranging
LSTM	Long Short-Term Memory
MLP	Multilayer Perceptron
nDTW	Normalized Dynamic Time Warping
NLP	Natural Language Processing
R2R	Room-to-Room
REC	Referring Expression Comprehension
RIS	Referring Image Segmentation
RNR	Referring Navigable Regions
SLAM	Simultaneous Localization and Mapping
TC	Task Completion
ViT	Vision Transformer
VLN	Vision-and-Language Navigation
X3D	Expand 3D (video CNN)

Chapter 1

Introduction

1.1 Scope of the Thesis

The focus of this thesis is Vision-and-Language Navigation (VLN) for autonomous driving in outdoor settings. Humans routinely navigate from linguistic descriptions of the environment, associating visual elements with words such as “the black car on the left” or “the bus stand after the traffic light.” Transferring this capability to an autonomous vehicle requires grounding language in a continuously changing visual scene and converting that grounding into actionable control.

Existing VLN methods fall short in this setting. Indoor approaches often assume a known map that can be discretized into a navigation graph [1, 2]. Outdoor methods frequently restrict the agent to a small discrete action set on pre-recorded street views [3, 4]. Single-image visual grounding methods for navigable regions [5] assume that the referred region is always visible in the current frame. None of these assumptions hold for live outdoor driving, where the environment is unknown, dynamic, and where commands routinely refer to regions that are not yet in view.

This thesis reformulates the Referring Navigable Regions (RNR) task for dynamic outdoor environments: at each timestamp the model predicts a segmentation mask corresponding to an intermediate or final navigable region, which is then passed to a local planner for continuous control. We also introduce CARLA-NAV, a meta-dataset for training and live evaluation in the CARLA simulator [6].

1.2 Research Problems

A *Grounding navigable regions when the referred goal is not visible.*

Single-image grounding methods [5, 7, 8] fail when the region described by the command is occluded, out of frame, or not yet visible. Everyday commands such as “take a right

once you see the traffic signal” require predicting intermediate navigable regions until the final goal comes into view.

B *Moving beyond discrete action spaces for outdoor VLN.*

Discretizing actions (FORWARD, LEFT, RIGHT, STOP) or environment graphs limits fine maneuvers such as “park between the two cars on the right” [3, 9]. Outdoor VLN needs continuous control over any drivable region on the road.

C *Exploiting temporal and trajectory context for live navigation.*

Static-image methods lack short-term and long-term temporal context. For multi-step commands (e.g., “turn left and park near the blue dustbin”), the model must know which part of the instruction has already been executed. Historical trajectory context and multi-frame visual features are essential for live inference in a changing scene.

1.3 Motivation

1.3.1 Limitations of Single-Image Visual Grounding

Referring Expression Comprehension (REC) [8, 10] and Referring Image Segmentation (RIS) [7] localize objects from language, but object boxes or masks do not answer “which region on the road to drive to.” Referring Navigable Regions (RNR) [5] addresses this by predicting a navigable region on the road for a linguistic command. However, RNR is limited to static images from pre-recorded sequences and assumes the target is always visible. As shown in Figure 1.1, the model correctly grounds a visible black car, yet completely confuses a left-turn command that does not exist in the current view.

1.3.2 Limitations of Graph-Based and Discrete-Action VLN

Indoor VLN often assumes a known topology and models navigation as graph traversal [1, 11, 12]. Outdoor datasets such as Touchdown [3] operate on Google Street View with four discrete actions. These designs preclude fine-grained outdoor maneuvers and make live evaluation in a fully navigable dynamic environment difficult. Prior CARLA-based language work has also been limited in action diversity [13], and Talk2Car focuses on localizing referred objects rather than navigating to regions [14].

1.3.3 Need for a Live Outdoor Benchmark

Evaluating on frames from pre-recorded videos does not measure whether an agent can complete an entire episode from start to finish in a changing scene. A dataset that supports



Figure 1.1: This is exactly the failure mode targeted by Problem A: single-image grounding succeeds when the referent is visible (black car, middle) but collapses when the command names a maneuver that is not yet in view (left turn). Example from the RNR model [5] on an image from their dataset. The thesis therefore predicts intermediate navigable regions until the final goal appears.

both supervised training on pre-recorded sequences and live validation in simulation is required to study language-guided outdoor navigation end to end.

1.4 Contributions

This thesis makes the following contributions:

1. We present CARLA-NAV, a vision-language navigation toolkit and meta-dataset on the CARLA simulator that enables fine-grained, language-based navigational maneuvers with continuous control via a local planner.
2. We propose a multi-task network for per-frame navigable-region prediction and short-term future trajectory prediction in dynamic outdoor environments. Both outputs are interpretable segmentation masks. We also document preliminary ViT and Conv3D (X3D) architectures that preceded the CLIP-based design. The network is demonstrated in Chapter 4; the ablation isolating the trajectory head remains future work (Section 4.10).
3. We perform real-time (live) navigation in CARLA with a diverse set of linguistic commands, using predicted masks as targets for the local planner and an area-based stopping criterion.
4. We provide extensive qualitative and quantitative ablations showing that multiple frames and historical contextual trajectory are both necessary for effective outdoor VLN, includ-

ing stratified displacement analyses for the earlier model lineage and formal analysis of grounding-first control.

Where each contribution is demonstrated:

- Contribution 1: Chapter 3 (collection methodology, Table 3.1) and Table 2.2 in Chapter 2.
- Contribution 2: Section 4.4 and Figure 4.1 in Chapter 4.
- Contribution 3: Section 4.4.4, Algorithm 1, and Table 4.1 in Chapter 4.
- Contribution 4: Tables 4.4, 4.5, and 4.6–4.9, plus the formal analysis in Section 4.3.

1.5 Thesis Layout

- C1 This chapter introduces the scope of the thesis, the research problems, motivation from prior work, and the main contributions.
- C2 Chapter 2 reviews background on autonomous navigation, computer vision, natural language processing, vision–language embeddings, language-guided navigation datasets and methods, and the CARLA simulator.
- C3 Chapter 3 describes the CARLA-NAV data collection process, dataset characteristics, and challenges encountered during curation.
- C4 Chapter 4 presents the problem formulation, a formal analysis of grounding-first control, the multi-task network, live-navigation pipeline, evaluation metrics, CLIP-family results, preliminary ViT/Conv3D architectures with stratified ablations, and an extended evaluation protocol.
- C5 Chapter 5 concludes the thesis and outlines directions for future work.

Chapter 2

Background

In this chapter, we review the concepts that underpin the rest of the thesis, with emphasis on a critical rather than textbook survey of Vision-and-Language Navigation (VLN). We briefly situate autonomous navigation, computer vision, and language understanding as they relate to language-guided driving; discuss vision–language embeddings with a focus on CLIP and dense adaptations; then taxonomize VLN methods, compare families by what fails under outdoor dynamic driving, and map the resulting research gaps to Problems A–C from Chapter 1. We close with the CARLA simulator and a summary linking this background to Chapters 3 and 4.

2.1 Introduction to Autonomous Navigation

Before specializing to language-guided driving, we recall the classical navigation pipeline so that the role of our grounding network is clear. Autonomous navigation is the science of enabling a robot or a vehicle to move through its environment without continuous human control [15]. It is fundamental to self-driving cars, delivery robots, and related systems [16, 17]. The classical pipeline comprises several interconnected stages: perception, localization, mapping, path planning, and control [18].

For road vehicles, the importance of autonomous navigation can be summarized as follows:

1. **Safety and efficiency:** Reducing human error while improving traffic flow and mobility [19].
2. **Real-time operation:** Perception and planning must run at rates compatible with driving dynamics [20].
3. **Coexistence with humans:** Agents must interact safely with human-driven vehicles, pedestrians, and cyclists in mixed traffic [19].
4. **Open research problems:** Dynamic scenes, language-conditioned goals, and continuous control remain active research areas that motivate the work in this thesis.

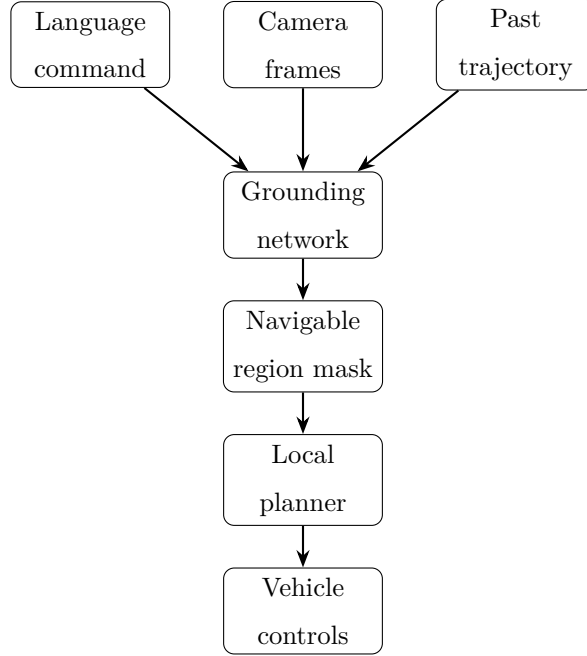


Figure 2.1: High-level Vision-and-Language Navigation (VLN) setup in this thesis. Language and visual observations are grounded into a navigable-region mask, which a local planner executes in continuous action space. This factorization is the structural break from nav-graph and discrete-action families surveyed below: the vehicle can be asked to drive toward any point on the road, not merely to choose from a fixed action menu.

Perception uses sensors such as cameras, LiDAR, radar, and ultrasonics to observe the environment [21, 16]. **Localization** estimates the vehicle’s pose within that environment [15], using odometry, inertial measurement units (IMUs), GPS, or Simultaneous Localization and Mapping (SLAM) [22]. **Mapping** builds a representation of the surroundings, from a simple 2D occupancy grid to a rich 3D map, for planning and obstacle avoidance [23, 24]. **Path planning** computes a route from the current pose to a goal while accounting for obstacles [25, 26]. **Control** executes the plan by issuing actuator commands and correcting deviations from feedback [27].

Common navigation paradigms include:

- **Reactive navigation:** real-time responses without a global map [28, 29]; fast in dynamic scenes, but not always globally efficient.
- **Map-based navigation:** planning on a known map [15, 24]; efficient when the map is accurate, brittle when the world changes.

- **SLAM:** jointly estimating pose and map in an unknown environment [22, 30, 31], with LiDAR-based and visual variants [32].

In this thesis, the local planner plays the role of path planning and control, while our contribution focuses on perception and language understanding: predicting which road region the vehicle should drive toward given a linguistic command (Figure 2.1).

2.2 Computer Vision for Language-Guided Driving

With the navigation pipeline in place, we next isolate the vision capabilities that language-guided driving actually needs. For this thesis, computer vision matters insofar as it supplies dense spatial predictions that can be aligned with language [33, 34]. Object detection localizes named entities [35, 36]; semantic segmentation labels every pixel with a class such as road or vehicle [37, 38]. Referring Image Segmentation goes further by conditioning the mask on a phrase [7]. Our navigable-region task is closest to the last: the network outputs a pixel-level mask of the region to drive toward, not a discrete action and not merely a box around a referred object.

2.2.1 Vision Encoders in Autonomous Navigation

Convolutional Neural Networks (CNNs) are widely used for visual recognition [39], including ResNets [40], DenseNets [41], EfficientNets [42], and VGG-style backbones [43]. In VLN, such encoders are often paired with language models such as Long Short-Term Memory (LSTM) networks [44] to follow instructions [1]. Vision Transformers (ViTs) treat an image as a sequence of patches and apply self-attention for more global context [45, 46].

Krantz et al. [2], in *Beyond the Nav-Graph: Vision-and-Language Navigation in Continuous Environments*, use a ResNet-50 visual encoder and argue that discrete navigation-graph assumptions can inflate reported performance. Controllable Imitative Reinforcement Learning (CIRL) similarly relies on vision encoders to learn driving policies from visual input [47]. Imaging laser radars have also been studied as complementary sensors for robotics perception [48].

For this thesis, the key takeaway is that strong visual features, and especially features that can be aligned with language, are a prerequisite for grounding navigable regions. Chapter 4 therefore uses CLIP visual features rather than a purely vision-only CNN backbone.

2.3 Language Understanding for Navigation

Vision alone does not specify the goal; language must be mapped to actionable structure in the scene [49, 1]. Modern systems rely on contextual encoders such as BERT [50] and on attention over multimodal tokens [46, 51]. Outdoor driving commands in this thesis are short

but referentially rich, covering objects, colors, spatial relations, and maneuvers, so the encoder must preserve word-level distinctions and align them with visual features. Raychaudhuri et al. [52] emphasize language-aligned supervision for tracking instruction progress; DUET [53] combines coarse map reasoning with fine grounding indoors. We instead use a jointly pretrained vision–language encoder (CLIP) so that command and road image live in one embedding space before dense prediction, without assuming an indoor map.

2.4 Combining Vision and Language: Vision–Language Embeddings

Having treated vision and language separately, we now turn to embeddings that place them in one space. Vision–language embeddings place images and text in a common representation space so that semantic matches lie close together [54, 55].

2.4.1 Introduction to CLIP and Beyond

CLIP (Contrastive Language–Image Pretraining) [56] is trained on large-scale internet image–text pairs. An image encoder and a text encoder are jointly optimized so that matched pairs are close in a shared embedding space (Figure 2.2). This dual-encoder design makes CLIP a natural backbone for VLN, where camera frames and linguistic commands must be aligned.

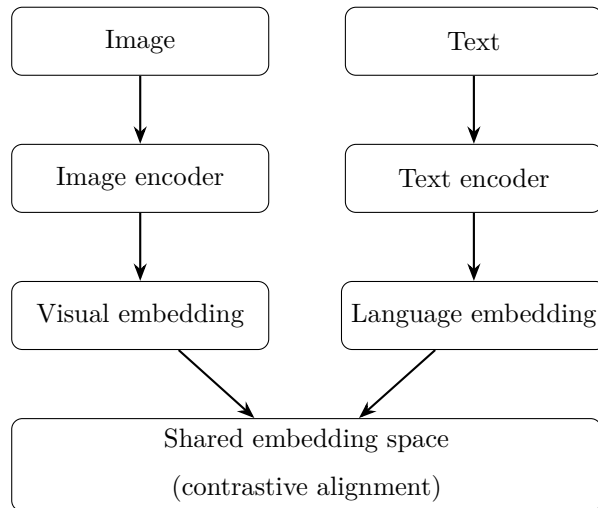


Figure 2.2: CLIP [56] encodes images and text into a shared embedding space via contrastive learning. Because command and image already share one space, the fusion block in Chapter 4 starts from an aligned representation rather than learning alignment from only 500 training episodes.

Larger generative models such as GPT-4 [57] and multi-modal architectures such as XDecoder [58] further expand the space of vision–language systems. They are not used in this thesis, but they illustrate the broader trajectory of the field toward richer multimodal reasoning.

2.4.2 Why They Matter for Language-Guided Navigation

In language-guided navigation, vision–language embeddings let a system relate what it sees to what it is told to do [3]. Dense-prediction adaptations of CLIP are especially relevant to our mask outputs: DenseCLIP converts image–text matching into pixel–text matching [59], and CLIPSeg shows that frozen CLIP features can be decoded into referring segmentations from text prompts [60]. Both lines of work support treating VLN as dense grounding rather than as action classification.

The practical implication for Chapters 3 and 4 is architectural: we treat VLN as grounding a command into a navigable region using CLIP features, then hand that region to a planner, rather than translating language directly into a discrete action. Chapter 4 further explains why we migrated from earlier GloVe+LSTM and X3D/ViT stacks to CLIP as the multimodal backbone.

2.5 Language-Guided Navigation

We now taxonomize prior VLN work along the axes that determine whether an approach can transfer to dynamic outdoor driving. Language-guided navigation develops systems that interpret natural language instructions and use them to move through a physical or virtual environment [1]. It combines NLP, computer vision, and sequential decision-making. Challenges that matter for outdoor driving include referential ambiguity (the same words can denote different road regions), linguistic variability, reliable multimodal fusion under changing traffic, and control rates compatible with vehicle dynamics [20, 61]. Applications include passenger voice commands in autonomous vehicles and interactive robots; our focus is the hard outdoor case of grounding *intermediate* navigable regions when the final referred goal is not yet visible.

2.5.1 Overview

Rather than listing methods alphabetically, we organize the literature along five axes that determine whether an approach can transfer to dynamic outdoor driving (Table 2.1):

1. **Environment representation:** navigation graph (nodes/edges) versus continuous free space.
2. **Action space:** discrete symbolic actions versus continuous controls or planner goals.

3. **Supervision:** imitation / behavior cloning, reinforcement learning, or dense mask / region labels.
4. **Fusion strategy:** late fusion of separately encoded modalities, early concatenation, or cross-/self-attention over joint tokens.
5. **Evaluation protocol:** scoring offline frames or panoramic hops versus live closed-loop episodes.

Family / exemplar	Env. repr.	Action	Supervision	Fusion	Eval.
R2R seq2seq [1]	Nav-graph	Discrete	IL+RL	Late / attn	Offline hops
RCM / EnvDrop [62, 63]	Nav-graph	Discrete	RL / aug.	Cross-modal	Offline hops
VLN-BERT / HAMT [51, 64]	Nav-graph	Discrete	IL	Transformer	Offline hops
VLN-CE [2]	Continuous	Low-level	IL	CNN+LSTM	Continuous sim.
E.T. [65]	Continuous	Low-level	IL	Transformer	Continuous sim.
Touchdown agents [3, 4]	Street graph	Discrete	IL	Cross-modal	Offline panoramas
Talk2Car [14]	Image	Object box	Dense / det.	Multimodal	Offline frames
RNR [5]	Image	Region mask	Dense mask	Multimodal	Offline frames
Grounding-first (ours)	Continuous road	Planner goal	Dense mask	CLIP+attn	Live CARLA

Table 2.1: Taxonomy of representative VLN and grounding methods along axes relevant to outdoor driving. Reading left to right shows why most prior families cannot transfer: they assume a nav-graph or discrete actions, use offline hop evaluation, or ground objects rather than navigable regions. “IL” denotes imitation learning; “hops” denotes discrete panoramic transitions.

2.5.2 Existing Datasets

The taxonomy above is incomplete without asking which datasets can support outdoor continuous control. Several datasets have shaped the field. Anderson et al. [1] introduce Room-to-Room (R2R) over Matterport3D indoor panoramas with long natural-language instructions. Outdoor Street View settings include Touchdown [3] and StreetNav [66], which support discrete panoramic navigation rather than continuous driving. Interactive indoor simulators such as AI2-THOR [11] and House3D [67], and RGB-D resources such as AVD [68] and 2D-3D-S [69], underpin much indoor work but do not model road traffic. Mapillary sequences [70, 71] provide street-level imagery for recognition and segmentation, not instruction following. Closer to our setting, Talk2Car [14] grounds referred objects in nuScenes driving scenes, and Referring Navigable Regions (RNR) [5] predicts navigable road masks on static frames.

As shown in Table 2.2, the datasets most relevant to outdoor language-guided driving differ sharply on dynamics, action space, and live evaluation.

Dataset	Environment	Dynamics	Action space	Annotation	Live eval.
R2R [1]	Indoor	Static scenes	Graph nodes	Path + instr.	No
Touchdown [3]	Outdoor	Static panoramas	Discrete actions	Path + instr.	No
StreetNav [66]	Outdoor	Static panoramas	Discrete actions	Path + instr.	No
Talk2Car [14]	Outdoor	Dynamic (nuScenes)	Object grounding	Object box	No
RNR [5]	Outdoor	Static frames	Region grounding	Navigable mask	No
CARLA-NAV (ours)	Outdoor	Dynamic traffic	Continuous (planner)	Navigable mask	Yes

Table 2.2: Comparison of language-guided navigation and grounding datasets along axes relevant to outdoor autonomous driving. CARLA-NAV (Contribution 1) uniquely combines outdoor dynamic scenes, continuous control via a local planner, navigable-region masks, and live episode-level evaluation. Details are given in Chapter 3.

2.5.3 Critical Comparison of Method Families

Nav-graph indoor VLN. Sequence-to-sequence agents [1], reinforced cross-modal matching [62], progress monitoring [72, 73], EnvDrop [63], and transformer policies such as VLN-BERT and HAMT [51, 64] achieve strong results when the environment is a known connectivity graph. What breaks outdoors is the assumption itself: a static nav-graph is unavailable for open roads, and graph hops cannot express fine parking or lane-placement maneuvers. Krantz et al. [2] further argue that nav-graph evaluation can inflate success by constraining the agent to oracle navigable nodes.

Continuous indoor agents (VLN-CE). VLN in continuous environments [2, 65, 74] removes the graph but typically still targets indoor locomotion with low-level move/turn actions. They do not address multi-agent road traffic, vehicle-scale kinematics, or language that refers to parking slots between cars.

Discrete outdoor Street View VLN. Touchdown-style agents [3, 4, 9] operate on pre-recorded panoramas with a handful of actions. Vision can be surprisingly weak as a cue when route structure alone succeeds [75, 4]. For driving, the discrete alphabet cannot represent “park between the two cars,” and offline panoramic hops are not live closed-loop control under dynamic agents.

Object grounding in driving scenes. Talk2Car [14] localizes referred objects but does not answer which road region to drive to, and evaluation is offline. Object boxes are insufficient when the command denotes a free region rather than an entity.

Static navigable-region grounding. RNR [5] predicts a road mask for a command, the right output type for our setting, but assumes the referred region is visible in a single static frame from a pre-recorded video. It fails on deferred goals (“turn right once you see the signal”), lacks temporal and trajectory context, and is never validated on full live episodes.

End-to-end driving with language. CARLA language work such as Talk2Nav-style way-point selection [13] and classical end-to-end driving [61, 76] either restrict maneuver diversity or bury language in opaque control outputs. Modular and hybrid pipelines [77, 78] improve interpretability but still need a grounding interface suited to continuous outdoor control.

2.5.4 Research Gaps and Thesis Mapping

The survey above leaves three concrete gaps, which map onto the research problems stated in Chapter 1:

Gap A → Problem A

Deferred and partially observable goals. Static grounding (RNR, Talk2Car) assumes the referent is in the current frame. Outdoor commands routinely refer to regions that are occluded or not yet visible. **Need:** intermediate navigable-region prediction over time.

Gap B → Problem B

Action-space mismatch. Nav-graph and Street View agents use discrete actions whose reachable set cannot express fine outdoor maneuvers. **Need:** continuous control via grounded regions passed to a local planner.

Gap C → Problem C

Missing temporal and progress context for live evaluation. Offline frame scoring does not measure episode completion under dynamic traffic; single-image models lack multi-frame and trajectory context for multi-step commands. **Need:** multi-frame features, historical trajectory context, and live CARLA evaluation.

Chapters 3 and 4 address these gaps with CARLA-NAV and a grounding-first network. Dense CLIP-based prediction [59, 60] supplies the representation; DUET-style coarse/fine reasoning [53] and language-aligned recovery ideas [52] motivate keeping explicit progress context (our trajectory map), adapted here to outdoor driving rather than indoor graphs.

2.6 Introduction to CARLA Simulator

The gaps above require a simulator in which nav-graph and discrete Street View assumptions can be relaxed. CARLA (Car Learning to Act) is an open-source simulator for autonomous driving research [6]. It provides urban layouts, vehicles, pedestrians, configurable weather and sensors, and an API for controlling the simulation. A server–multi-client architecture supports scalable experiments. Sensors include cameras, LiDAR, depth, and GPS. Maps follow the OpenDRIVE standard; ScenarioRunner defines traffic situations; a ROS bridge is available. CARLA also ships baseline agents and supports record/replay for controlled comparisons.

These properties make CARLA suitable for CARLA-NAV: we can spawn dynamic traffic, collect synchronized camera and pose data, project clicked navigable regions into masks, and later evaluate live navigation rather than only scoring pre-recorded frames.

2.7 Summary

This chapter reviewed autonomous navigation, the vision and language building blocks for grounding, and critically a taxonomy of VLN methods along environment representation, action space, supervision, fusion, and evaluation protocol (Table 2.1). Dataset comparison (Table 2.2) and the family-by-family critique show that prior work leaves deferred outdoor goals, continuous fine maneuvers, and live dynamic evaluation under-addressed. Section 2.5.4 maps those gaps to research problems A–C. CARLA provides the simulator in which the assumptions of nav-graphs and discrete Street View actions can be relaxed. Chapter 3 introduces CARLA-NAV; Chapter 4 presents the formal grounding-first analysis, multi-task network, and live-navigation results.

Chapter 3

CARLA-NAV Dataset

Having established the limitations of existing VLN methods in Chapters 1 and 2, we now describe the dataset designed to overcome those limitations. The quality and relevance of the data collected directly influence the performance and effectiveness of the final model; in autonomous driving this is especially true because of the complexity and variability of real-world driving scenarios.

In this chapter, we describe the creation of CARLA-NAV (Contribution 1), the meta-dataset that supports the language-guided navigation approach developed in this thesis.¹ We began with Grand Theft Auto V (GTA V) as an initial data collection environment, then switched to the research-oriented CARLA simulator [6]. We also explain our shift from predicting steering, acceleration, and brake values to predicting segmentation maps for navigation, detail the observer–navigator–verifier annotation protocol, and summarize dataset statistics and challenges.

3.1 Introduction

Chapter 1 motivated the need for a live outdoor benchmark that supports deferred goals, continuous control, and episode-level evaluation; Chapter 2 mapped those needs to research gaps A–C. CARLA-NAV is the dataset response: outdoor dynamic scenes with dense navigable-region and trajectory annotations, continuous control via a local planner, and live evaluation in the CARLA simulator [6]. Combined with a visual grounding-based approach, the dataset allows fine-grained control over the vehicle by enabling navigation to any drivable region on the road. Live language-guided navigation in simulation has prior precedents; our contribution is a continuous-control, region-grounding formulation with dense annotations for outdoor dynamic scenes.

¹Parts of this chapter appear in Jain et al., *Ground then Navigate: Language-guided Navigation in Dynamic Scenes*, IEEE ICRA 2023 [79].

3.2 Initial Data Collection with GTA V

Before settling on CARLA, we first attempted collection in a commercial game engine to test whether language-conditioned driving data could be obtained at all. We used Grand Theft Auto V (GTA V) and the DeepGTAV and V-Pilot mods. The game’s realistic graphics, diverse urban environments, and dynamic traffic made it an appealing choice for simulating real-world driving conditions. DeepGTAV allows extraction of camera images, vehicle state, and information about other agents in the scene.

Our initial approach was to train a deep learning model to predict steering, acceleration, and brake. However, we encountered several challenges. First, predicting only three continuous controls without sufficient high-level context proved brittle, consistent with observations that end-to-end driving models can struggle without structured guidance [80, 61]. Second, supervised learning alone was insufficient without a carefully designed reward if we moved to reinforcement learning, and defining rewards for language-conditioned driving is itself a hard problem [81, 82]. GTA V also did not provide the per-pixel and camera-calibration access needed for dense navigable-region annotation.

Given these challenges, we switched to predicting segmentation masks under supervised learning. Dense mask prediction provides a more interpretable target and a clearer visual representation of what the model believes is navigable [83, 84]. In the next section, we discuss why CARLA made this approach feasible.

3.3 Switching to CARLA for Data Collection

The GTA V limitations above motivated a move to a research simulator with open calibration and environment state. After encountering those limitations, we switched to CARLA [6], an open-source simulator designed for autonomous driving research. CARLA offers several advantages that made it suitable for our purposes:

1. **Research-oriented control:** CARLA provides flexible control over weather, time of day, and the behavior of other vehicles and pedestrians, which is crucial for a diverse dataset.
2. **Consistent physics:** Vehicle motion and interactions with the environment are more reliable than in a commercial game pipeline.
3. **Rich environment state:** Access to 3D object positions, road layout, and sensor meta-data supports precise annotation.
4. **Camera calibration:** Live camera intrinsics and orientation matrices enable transforming 2D screen coordinates into 3D world coordinates, essential for projecting navigable regions and driving a local planner.

These capabilities allowed us to adopt visual grounding: associating linguistic descriptions of navigable regions with their visual counterparts and predicting a segmentation map that can be used for navigation. This shift would not have been feasible in GTA V due to limited access to internal game data.

3.4 Data Collection Methodology

With CARLA available, we next define how episodes are authored so that only language and clicks require human effort. The data collection process in CARLA was a collaborative effort involving three roles: the observer, the navigator, and the verifier.

- **Observer:** Provides clear linguistic commands based on the current environment state. Commands are generated on-the-fly rather than scripted, capturing natural variation.
- **Navigator:** Drives in CARLA by clicking navigable regions in the front-camera view and follows the observer’s command as naturally as possible. A restart option is available for erroneous clicks.
- **Verifier:** Reviews the language command, navigated path, and generated masks, and decides whether the episode is included in the dataset.

The 2D screen coordinates clicked by the navigator are transformed into 3D world coordinates using the inverse projective transform with CARLA’s camera intrinsic and orientation matrices under a flat-ground assumption. This 3D position is passed to the local planner (CARLA’s default planner in our case; the design is modular and can later use planners such as NEAT [85]). Early collection prototypes also experimented with reading depth from a CARLA depth camera for the clicked pixel; the released CARLA-NAV annotations follow the flat-ground rectangle projection documented below and in the archival paper.

An episode comprises multiple mouse clicks until the final navigable region becomes visible in the front view. Intermediate clicks signify intermediate navigable regions; the last click depicts the final goal region. Mouse clicks are converted into segmentation masks by drawing a $3\text{m} \times 4\text{m}$ rectangle (approximately the size of a car) in the top view, centered at the click, and projecting that rectangle into the front camera. Overall, only language commands and mouse clicks require manual effort; the rest of the annotation process is automated. Early drafts treated intermediate and final regions as two separate segmentation classes to drive stopping; the released dataset and CLIP-family models use a single navigable-region mask, with the area-based stopping rule described in Chapter 4.

For the trajectory prediction task used later for interpretability, we take the 3D position of the vehicle in successive frames and project those positions onto the front-camera image. Trajectory prediction is treated as a dense prediction task and is not quantitatively evaluated.

As shown in Figure 3.1, the collection loop automates everything after the click; Figure 3.2 shows ground-truth annotations for three sampled frames from an episode of CARLA-NAV.

3.5 Dataset Characteristics

The methodology above yields episodic data whose statistics and diversity we now report. Relative to prior language-guided navigation and grounding datasets surveyed in Chapter 2 (Table 2.2), CARLA-NAV is designed for outdoor dynamic scenes with continuous control and live episode-level evaluation. CARLA-NAV contains episodic data: each episode consists of a language command and the corresponding CARLA video of navigation toward the final goal region described by the command. The ground-truth segmentation mask for each frame corresponds to either the final or an intermediate navigable region. Each frame is additionally annotated with a plausible future trajectory of the vehicle in the next few frames.

The dataset includes video sequences captured in eight different maps, fourteen distinct weather conditions, and a diverse range of dynamically moving vehicles and pedestrians. Language commands contain detailed visual descriptions and describe a wide range of maneuvers. Commands can have a single maneuver (e.g., park behind the black car on the left) or multiple maneuvers (e.g., take a right turn and park near the bus stand), with a maximum of three maneuvers.

Table 3.1 reports core episode statistics. Since each episode contains multiple frames, the overall dataset size is 83,297 frames across all splits. During training we use pre-recorded sequences. During inference on validation and test splits, the vehicle is spawned at the corresponding starting location and navigation is performed live based on network prediction, not on the pre-recorded sequences.

The split sizes evolved during the project: an early draft used 441 train / 25 validation episodes; an intermediate draft used 500 train / 50 validation (with an internal note to split validation into 25/25); the released and archival configuration is 500 / 25 / 34 train/val/test as in Table 3.1. The additional nine test episodes relative to a 25/25 split were added before the CLIP-family evaluation reported in Chapter 4.

3.5.1 Language and Maneuver Diversity

Commands mainly consist of up to three subcommands, most having two. Subcommands include turning, going straight, stopping, and parking, and reference pedestrians, infrastructure, road elements, other vehicles, and scenery. A manual audit of the command list yields on the order of 10^2 unique tokens and a few hundred distinct command strings after lowercasing (many near-paraphrases); we do not claim an exact vocabulary size without a released tokenizer dump. Relative to indoor VLN datasets such as R2R [1], whose instructions average on the order of

Split	# Episodes	# Frames	Cmd len.	Clicks	Dist. (m)
Train	500	75,010	6.92	1.94	34.18
Val	25	3,300	6.76	2.00	37.26
Test	34	4,987	7.44	1.97	34.71

Table 3.1: Dataset statistics for CARLA-NAV (Contribution 1). Command length, clicks, and distance are averages per episode. The small validation and test sizes (25 / 34) set the granularity of every Task Completion claim in Chapter 4.

29 words, CARLA-NAV commands are short (6.9–7.4 words). The language task is therefore substantially simpler than long-horizon indoor instruction following; the difficulty instead lies in dynamic outdoor grounding and continuous control. Chapter 4 reports stratified displacement metrics for earlier ViT/Conv3D models under the same target, action, and subcommand taxonomies, and specifies a TC-stratified per-maneuver protocol for the CLIP family.

Held-out episode commands (validation $n = 25$, test $n = 34$) have been labeled under those same Chapter 4 taxonomies: referred target type, primary action verb, and subcommand count. Tables 3.2–3.4 report the counts. These tables describe the held-out splits only. The labeled listing is archived at `figures/vln/carla_nav_val_test_commands.csv`.

Target type	Val	Test
infrastructure	10	11
road/light	13	20
automobile	2	3
Total	25	34

Table 3.2: Referred target-type counts on the CARLA-NAV held-out splits ($n_{\text{val}} = 25$, $n_{\text{test}} = 34$), using the same taxonomy as Table 4.8. Training commands are not yet included.

TODO: Add the train-split command list and extend Tables 3.2–3.4 with a Train column (and full-dataset totals). Held-out val/test counts above are complete.

Primary action	Val	Test
stop	5	6
go	3	4
take	4	3
turn	4	14
change	2	1
drive	4	3
park	3	3
Total	25	34

Table 3.3: Primary action-verb counts on the CARLA-NAV held-out splits ($n_{\text{val}} = 25$, $n_{\text{test}} = 34$), using the same taxonomy as Table 4.7. Near-synonyms are mapped as wait $\rightarrow$ go, halt $\rightarrow$ stop, shift $\rightarrow$ change. Training commands are not yet included.

3.5.2 Maps and Weather

Episodes span eight CARLA maps and fourteen weather presets (clear, cloudy, wet, sunset, and night variants). Under the default protocol, episodes are shuffled so that maps and weathers appear in all of train, validation, and test in roughly balanced proportions. This design maximizes diversity within each small split, but it is *not* a held-out-map protocol: layout memorization remains possible. The map-unseen evaluation contract is defined in Chapter 4.

3.5.3 Mask Annotation Coarseness

Ground-truth navigable regions are $3\text{m} \times 4\text{m}$ rectangles projected from mouse clicks under a flat-ground assumption. This is a coarse proxy for the true drivable surface: the rectangle may overlap curbs or undershoot wide parking gaps, and slope transitions introduce projective error (listed below). We treat these masks as supervision for learning *approximate* goal regions rather than pixel-perfect road segmentation, and interpret quantitative mask-conditioned metrics accordingly.

3.6 Data Collection Challenges

The statistics above hide several practical difficulties that shaped both annotation quality and evaluation design. We encountered several practical challenges during data collection:

# Subcommands	Val	Test
1	16	14
2	9	20
3	0	0
Total	25	34

Table 3.4: Subcommand-count distribution on the CARLA-NAV held-out splits ($n_{\text{val}} = 25$, $n_{\text{test}} = 34$), using the same taxonomy as Table 4.9. No three-subcommand episodes appear in these held-out lists. Training commands are not yet included.

1. **Asynchronous simulation:** CARLA operates asynchronously, so the simulator does not wait for annotation code to finish before advancing. Careful synchronization was required so that frames being annotated matched the displayed state.
2. **Slopes:** Transforming 2D screen coordinates into 3D world coordinates assumes a flat ground plane. Transitions from slopes to flat surfaces introduced inaccuracies that required special handling.
3. **Split contamination:** Given the finite CARLA maps, similar scenarios can appear across train, validation, and test splits. Shuffling episodes and randomizing spawn locations diversifies each split but *does not* hold out maps; a model may still exploit shared layout. Chapter 4 therefore defines a map-unseen protocol as the stricter evaluation target.
4. **Subjectivity:** Different observers and navigators may interpret the same command differently. Role rotation and verifier checks reduced individual bias.

3.7 Data Collection Team and Conclusion

Having described the protocol and its challenges, we close with authorship and a forward pointer to modeling. The CARLA-NAV dataset was curated collaboratively. Annotation roles (observer, navigator, verifier) were rotated among the candidate, Amogh Tiwari, and Kanishk Jain (co-first author of the corresponding ICRA publication [79]). KV Aditya Srivatsa assisted with early data-collection tooling. Dr. Vineet Gandhi provided more complex linguistic commands and acted as a verifier in ambiguous cases. A fuller contribution statement appears under Related Publications.

In conclusion, CARLA-NAV provides a diverse set of language-guided outdoor navigation episodes with dense navigable-region and trajectory annotations. The dataset and collection toolkit form the foundation for the multi-task network and live-navigation experiments presented in Chapter 4.

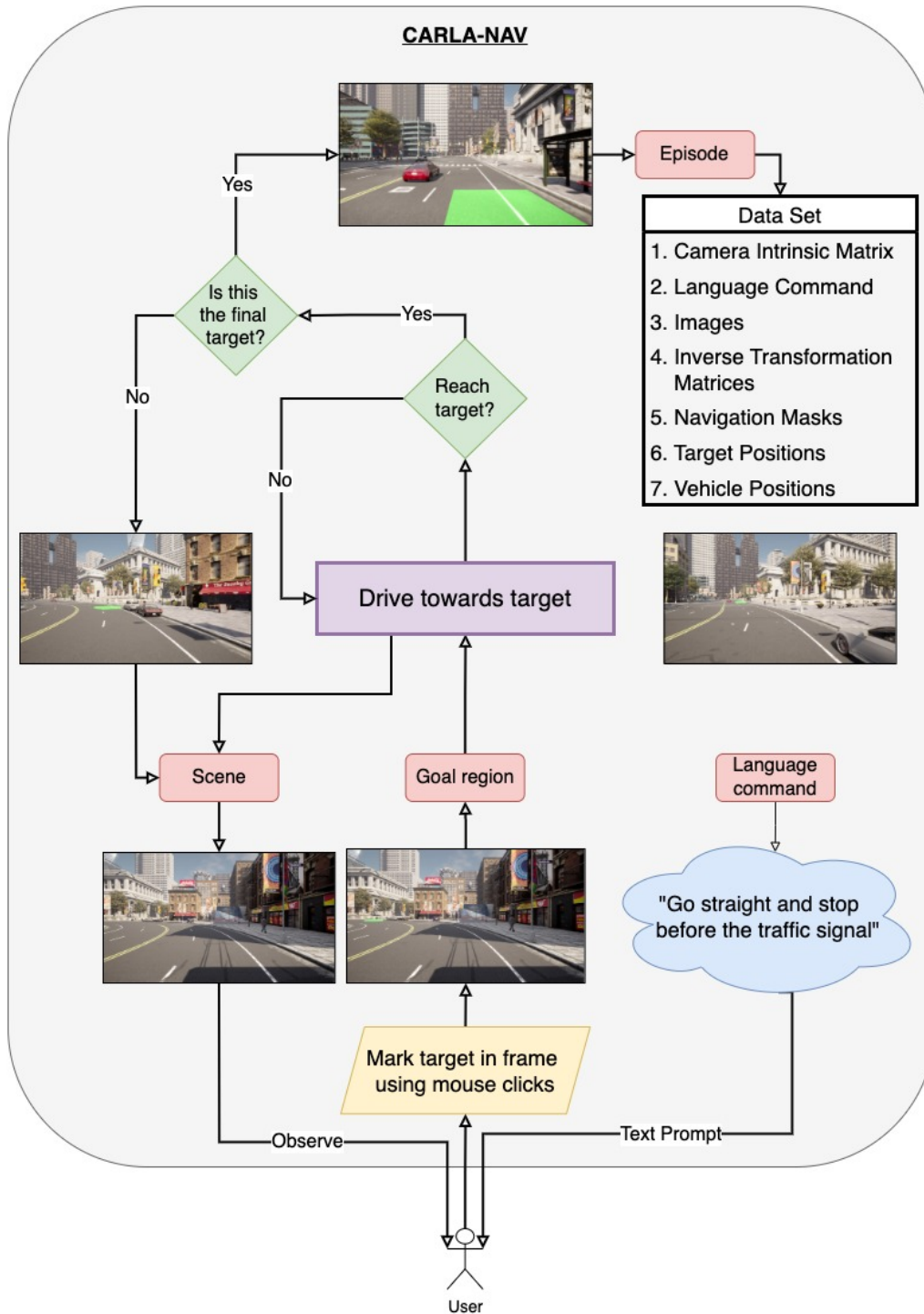


Figure 3.1: Pipeline illustrating how only language commands and mouse clicks require manual annotation, while all remaining supervision (masks, camera matrices, and trajectory targets) is generated automatically from CARLA’s calibration. The toolkit drives toward each clicked target until the final destination and stores the episode.

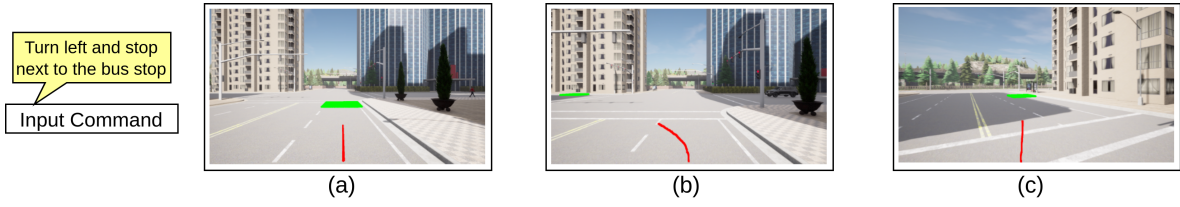


Figure 3.2: Ground-truth annotations of three sampled frames from a CARLA-NAV episode (command at top; green = navigable region, red = short future trajectory). Panel (a) is the key design point: at episode start neither the turn nor the final goal is visible, so supervision is an *intermediate* region. That is what makes deferred-goal training possible. (b) Intermediate mask for the left turn; (c) final region (stop next to the bus stop).

Chapter 4

Vision-and-Language Navigation in Autonomous Driving

This chapter presents our approach to Vision-and-Language Navigation (VLN) for autonomous driving in dynamic outdoor environments.¹ Building on the CARLA-NAV dataset introduced in Chapter 3, we formalize the problem, describe a multi-task grounding network, explain live navigation, and report quantitative and qualitative results. Dataset creation details are not repeated here; we focus on modeling, inference, and evaluation.

4.1 Background and Related Work

Chapter 2 already taxonomizes VLN families and maps gaps A–C; here we only add method-level citations for visual grounding that feed our formulation. Visual grounding associates linguistic descriptions with visual counterparts via proposal-based Referring Expression Comprehension (REC) [8, 10, 86, 87, 88, 89, 90, 91] or pixel-level Referring Image Segmentation (RIS) [92, 93, 94, 95]. Referring Navigable Regions (RNR) [5] localizes navigable road regions for language commands, but is limited to static images in pre-recorded sequences [96]. We reformulate RNR for dynamic outdoor settings and perform live navigation with continuous control via a local planner. Continuous-control grounding of free-form navigable regions remains comparatively rare relative to discrete-action outdoor VLN.

4.2 Problem Statement

Having situated the task against prior work, we now state the prediction problem the network must solve. Intuitively, the network continuously predicts the next drivable region rather than the final destination: the command says which region, and the past trajectory says how much of the command is already done.

¹This chapter is based on Jain et al., *Ground then Navigate: Language-guided Navigation in Dynamic Scenes*, IEEE ICRA 2023 [79].

Given video frames $V = \{v_{t-k}, v_{t-k+1}, \dots, v_t\}$, a contextual historical trajectory P , and a language command $L = \{l_1, l_2, \dots, l_N\}$, where t is the current timestamp, k is the historical window size, and N is the maximum number of words, the goal is to predict the navigable mask y_t and the future trajectory mask z_t for the current frame v_t .

The contextual trajectory P provides information about which part of a multi-step command has already been executed. For example, for “turn left and park near the blue dustbin,” P indicates whether the left turn has already been taken. The spatial location of the navigation mask should determine the trajectory path’s direction, and vice versa.

4.3 Formal Analysis

Before describing the architecture, we state why a dense grounding-first formulation is a better inductive bias for outdoor language-guided driving than selecting from a small discrete action set.

4.3.1 Why Segmentation Dominates Discrete Action Selection

Intuitively, with a fixed menu of four actions there are places on the road the vehicle simply cannot be asked to go, no matter how much data it sees. Let $\mathcal{A}_{\text{disc}} = \{a_1, \dots, a_K\}$ be a finite action alphabet (e.g., FORWARD, LEFT, RIGHT, STOP as in Touchdown-style outdoor VLN [3]). Executing a sequence drawn from $\mathcal{A}_{\text{disc}}$ induces a finite reachable set $\mathcal{R}_{\text{disc}}$ of vehicle poses relative to the current state. The set of drivable poses on the road, $\mathcal{R}_{\text{road}}$, is a continuum: any point on the free surface is in principle a legal target. The Hausdorff gap

$$\delta = \sup_{x \in \mathcal{R}_{\text{road}}} \inf_{y \in \mathcal{R}_{\text{disc}}} \|x - y\| \quad (4.1)$$

is an irreducible quantization error of discrete control. Commands such as “park between the two cars on the right” require placing the vehicle at a geometrically constrained pose that typically lies outside $\mathcal{R}_{\text{disc}}$ for small K . The failure is therefore *representational*, not statistical: no amount of additional discrete-action data recovers a maneuver that the action space cannot express. By contrast, predicting a navigable-region mask y_t and taking its centroid (or any point sampled from the mask) as a planner target yields a goal in continuous world coordinates, so the reachable set under a competent local planner approximates $\mathcal{R}_{\text{road}}$ up to planner fidelity rather than up to K .

4.3.2 Advantages of Grounding-First

Intuitively, the network decides *where* to go; the planner decides *how*. We factorize the policy as

$$P(\text{controls} \mid V, P, L) = P_{\text{plan}}(\text{controls} \mid y_t, \text{state}) P_{\theta}(y_t \mid V, P, L), \quad (4.2)$$

where P_θ is the learned grounding network and P_{plan} is a fixed local planner. Three consequences follow:

1. **Supervision density.** Each frame provides $O(HW)$ pixel-level targets for y_t , versus a single categorical label under action classification. Dense losses (Eq. 4.4) therefore supply far richer gradient signal per demonstration.
2. **Dynamics as a prior.** Vehicle kinematics and collision avoidance enter through P_{plan} rather than being relearned inside P_θ .
3. **Inspectability.** Every decision yields an explicit mask that can be visualized and overridden. This is the operational basis of the interpretability claim, not a post-hoc explanation of a black-box action logit.

The short-term trajectory head z_t is an auxiliary dense prediction that shares M with the navigation head; it does not enter P_{plan} in the current system, but correlates with y_t and provides a visual cue of the intended short-horizon path. Early project drafts independently noted that incorporating the predicted trajectory into the planner is future work, confirming that the published system uses z_t for interpretability rather than control.

4.3.3 Complexity of Multimodal Fusion

Multi-head self-attention in Eq. 4.3 operates over $S = (T+1) + N$ multimodal tokens at spatial resolution HW . With $T = 8$, $H = W = 7$, and $N \leq 20$, we have $HW = 49$ spatial sites and on the order of $S \cdot HW \approx 441$ tokens when language is broadcast spatially (or S tokens after spatial pooling, depending on implementation). Self-attention is quadratic in the attended length, so the dominant cost scales as $O(((T+1)HW)^2C)$ when attending over the visual-trajectory grid. Doubling the frame count therefore roughly quadruples attention FLOPs. This motivates stopping the frame ablation at $n = 8$ (Table 4.4) and predicting only every fifth frame at inference time. Trainable parameters are concentrated in the fusion block and two segmentation heads; the CLIP ViT-B/32 encoders contribute the majority of total parameters.

4.3.4 Convergence Intuition for Combo Loss

Navigable masks cover a small fraction of each frame, so the pixel prior is heavily skewed toward background. The BCE term L_{bce} then yields gradients dominated by easy negatives. The soft Dice coefficient $D(\hat{y}_t, y_t)$ is (approximately) scale-invariant to foreground size and directly optimizes overlap, counteracting that skew. Setting $\lambda = 0.3$ weights the objective toward Dice while retaining a calibrated probabilistic BCE term for stable early training [97]. Both heads share the same L_{combo} ; balancing them amounts to summing the two losses with

equal weight in our experiments. Severe class imbalance and the small absolute mask area also explain why area-based stopping is sensitive: area growth under perspective is a coarse proxy for goal proximity.

4.4 Methodology

The formal analysis above motivates a dense, multi-task architecture; we now specify its components. We propose a multi-task network for navigation-region prediction and future-trajectory prediction (Contribution 2). Both tasks are treated as dense prediction tasks for interpretability. During live inference, dense pixel points are converted to 3D world coordinates using inverse projective transformation. As shown in Figure 4.1, every output is a mask before the planner is invoked.

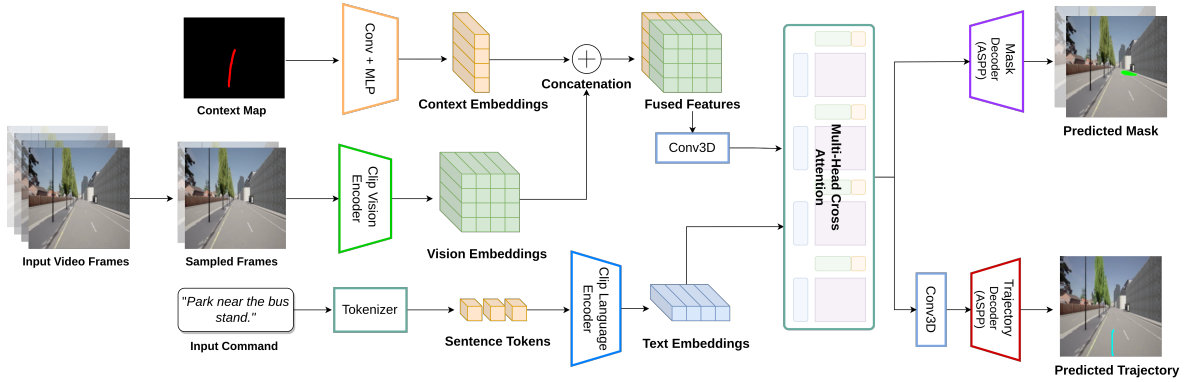


Figure 4.1: Overall pipeline of the proposed approach (Contribution 2). Given visual frames, the already executed past trajectory, and the textual command, the network predicts a navigable-region mask and a short-horizon trajectory mask. Every output is a mask, so every decision is inspectable before it reaches the planner.

4.4.1 Feature Extraction

We utilize CLIP [56] to extract linguistic and visual features. For the linguistic expression L , we tokenize the command with the CLIP tokenizer and obtain word-level features $F^l = \{f_1^l, \dots, f_N^l\}$ of shape $\mathbb{R}^{N \times C_l}$. For visual frames V , CLIP encodes video features $F^v = \{f_{t-k}^v, \dots, f_t^v\}$. For trajectory context P , we project the past trajectory onto an image of the same size as v_t and pass it through convolution and MLP layers to obtain F^p with the same feature size as the per-frame visual features.

4.4.2 Multi-Modal Fusion

Intuitively, we lay the frames, the past-trajectory image, and the words out as one sequence and let attention decide which words attend to which places and times. The trajectory context feature $F^p \in \mathbb{R}^{C_v \times 1 \times HW}$ is concatenated with video features F^v along the temporal dimension, yielding $F^{vp} \in \mathbb{R}^{C_v \times (T+1) \times HW}$. We then fuse F^{vp} with linguistic features $F^l \in \mathbb{R}^{N \times C_l}$ by length-wise concatenation along the sequence axis (denoted $[\cdot; \cdot]$ to avoid confusion with the Hadamard product), apply multi-head self-attention over the resulting multimodal tokens, and collapse the temporal axis with a 3D convolution:

$$\begin{aligned} F &= [F^{vp}; F^l] \in \mathbb{R}^{C \times S \times HW}, \\ A &= \text{Mhead}(F, F, F), \\ M &= \text{Conv3D}(AF) \in \mathbb{R}^{C_m \times H \times W}, \end{aligned} \tag{4.3}$$

where $S = (T+1) + N$ is the joint token length after concatenation, Mhead is multi-head self-attention over the multimodal sequence, AF denotes matrix multiplication of the attention weights with the value features, and Conv3D aggregates the temporal (and language) axis into a single spatial feature map M for the segmentation heads.

4.4.3 Segmentation Heads and Loss

Separate segmentation heads (sequences of convolutions with upsampling) predict the navigation and trajectory masks from M . We train with combo loss [97], a combination of pixel-wise binary cross-entropy and the Dice coefficient used as a soft overlap term. Intuitively, the mask is a small target in a big image, so a per-pixel loss can score well by predicting “not navigable” everywhere; Dice measures overlap and cannot be gamed that way (Section 4.3.4):

$$\begin{aligned} L_{\text{bce}} &= -\frac{1}{|\Omega|} \sum_{i \in \Omega} (y_{t,i} \log(\hat{y}_{t,i}) + (1 - y_{t,i}) \log(1 - \hat{y}_{t,i})), \\ D(\hat{y}_t, y_t) &= \frac{2 \sum_i \hat{y}_{t,i} y_{t,i}}{\sum_i \hat{y}_{t,i} + \sum_i y_{t,i} + \varepsilon}, \\ L_{\text{combo}} &= \lambda L_{\text{bce}} - (1 - \lambda) D(\hat{y}_t, y_t), \end{aligned} \tag{4.4}$$

where Ω indexes image pixels and ε is a small constant for numerical stability. We set $\lambda = 0.3$ so that the Dice term dominates. The same loss is applied to both heads; the approach is end-to-end trainable, and the predicted trajectory correlates with the predicted navigation mask as a short-term route cue.

4.4.4 Live Navigation

To convert predictions into closed-loop driving, we next specify how masks become planner targets (Contribution 3). We take the largest connected component and use its centroid as

the 3D goal (via inverse projective transform). As the vehicle approaches the final region, the predicted mask area tends to increase due to perspective; we therefore use an area-based rule to decide when to stop. Algorithm 1 summarizes the decision procedure used in the published CLIP-family experiments.

Algorithm 1 Target determination for live navigation (CLIP-family protocol)

Require: Predicted navigable mask $\mathbf{M}$, trajectory mask $\mathbf{T}$ (optional)

- 1: Threshold $\mathbf{M}$ and extract connected components
 - 2: Let C be the component with largest summed probability; $c \leftarrow$ weighted centroid of C
 - 3: **if** area of C exceeds an empirically chosen threshold for five consecutive predictions **then**
 - 4: Treat c as the final goal and stop navigation
 - 5: **else**
 - 6: Pass c to the local planner as the next intermediate target
 - 7: **end if**
-

The numeric area threshold was found empirically and was *not* recorded in the archival paper or earlier drafts; we therefore do not invent a value here. During inference we predict every 5 frames to reduce latency; training uses every frame of each episode.

Evolution of the stopping rule. The live-navigation rule evolved across project drafts, and documenting that evolution clarifies what the CLIP tables actually measure:

1. **Two-class mask with trajectory fallback.** Early prototypes predicted separate intermediate and final mask channels. The target was the weighted centroid of the dominant component; if neither channel exceeded a confidence threshold δ , the farthest point of the skeletonized trajectory mask was used instead. Three consecutive final predictions ended the episode.
2. **Two-band confidence counters.** An intermediate draft averaged the intermediate and final channels, applied $\delta_{\min}/\delta_{\max}$ bands, incremented a confident-prediction counter, and ended after a short no-new-target timeout.
3. **Single-mask area rule (reported).** The ICRA protocol, and all CLIP results in this chapter, uses a single navigable mask with five consecutive area triggers (Algorithm 1).

4.5 Evaluation Metrics

With the live pipeline defined, we next choose metrics that match the set-valued nature of language commands. Following prior VLN work [4, 98, 3], we use **Task Completion (TC)** as

the primary success metric: whether the agent successfully navigates to the goal location for a given language command. Importantly, TC in our evaluation is a *human* judgment of episode success rather than an automatic geometric check; inter-rater agreement was not recorded. We additionally use **Fréchet Distance** [99] and **normalized Dynamic Time Warping (nDTW)** [100] to compare the live predicted path with the ground-truth path.

Fréchet Distance and nDTW provide useful path-similarity signals, but they are not fully robust indicators of average navigation quality: a single early wrong turn can dominate the score, and a linguistic command may admit multiple valid trajectories (e.g., “park near the blue dustbin”). Task Completion therefore remains the primary metric, with the caveat that it inherits subjectivity from human labeling. Section 4.5.2 develops this construct-validity argument in full and motivates a reproducible goal-region success predicate (Section 4.10.4).

Because live CARLA evaluation is stochastic under dynamic traffic, we also report Wilson score 95% confidence intervals for TC (Table 4.2), computed from the binomial success counts implied by the point estimates on $n_{\text{val}} = 25$ and $n_{\text{test}} = 34$ episodes. These intervals quantify sampling uncertainty of a single evaluation pass under a binomial model; they do *not* capture annotator disagreement, and they are not a substitute for multi-seed re-runs (Section 4.10).

4.5.1 Displacement Metrics for Preliminary Models

Before adopting CLIP and Task Completion, earlier model variants were evaluated with trajectory-displacement metrics that remain useful for stratified analysis. Let $A = (a_1, \dots, a_{T_A})$ and $B = (b_1, \dots, b_{T_B})$ be predicted and ground-truth trajectories in world coordinates (CARLA meters).

- **Final Displacement Error (FDE):** L_2 distance between the terminal points, $\|a_{T_A} - b_{T_B}\|_2$.
- **Average Displacement Error (ADE):** mean L_2 distance after resampling both paths to a common length T , $\frac{1}{T} \sum_{i=1}^T \|a_i - b_i\|_2$.
- **ADE-Match:** because ADE requires equal-length trajectories, the preliminary evaluation used a coupling-based variant that replaces the Fréchet supremum with a mean over the coupling. Intuitively, this metric can be fooled: an agent that stalls in one place can score near zero. With continuous reparameterizations $\alpha, \beta : [0, 1] \rightarrow [0, 1]$,

$$\text{ADE-Match}(A, B) = \inf_{\alpha, \beta} \text{mean}_{t \in [0, 1]} d(A(\alpha(t)), B(\beta(t))).$$

This quantity is closely related to the *average* (integral) Fréchet Distance introduced by Buchin [101] and studied algorithmically by Maheshwari et al. [102]: the standard formulation integrates distances with respect to *arc length* and normalizes by curve length. The uniform-in- t infimum above is degenerate: a monotone reparameterization can linger

near one point of B for almost the whole interval and sweep the remainder over vanishing measure, so a stalled agent can approach zero. We report ADE-Match as computed in the preliminary ViT/Conv3D evaluation (Section 4.6.3) and interpret it with that caveat.

4.5.2 Construct Validity: Why Task Completion Is Primary

Intuitively, “park near the blue dustbin” does not name one path; it names a family of acceptable ones, and a metric that compares against one recorded path will punish the acceptable alternatives. A language command C does not designate a single trajectory. It designates a *set* $\Phi(C)$ of trajectories that satisfy the command. The recorded demonstration τ^* is one sample from $\Phi(C)$, chosen by the annotator on that day. Any distance $d(\hat{\tau}, \tau^*)$ is a valid measure of correctness only if small distance is equivalent to membership in $\Phi(C)$. For referential outdoor commands that equivalence fails in both directions, which is why the thesis promotes Task Completion over the displacement suite as the primary verdict.

Two error directions.

1. **Right path, wrong endpoint.** An agent may track the road closely and still stop short of, or beyond, the referred goal. ADE dilutes that terminal error by averaging it against a well-tracked prefix; ADE-Match dilutes it further under the degenerate coupling above. FDE captures the endpoint, but ignores whether intermediate constraints were met. The preliminary Conv3D numbers make this concrete (Section 4.6.3): Fréchet Distance and FDE means nearly coincide (17.84 vs. 17.83 m), so the Fréchet supremum is attained at the terminal point in essentially every episode; the modal failure is right-path/wrong-endpoint.
2. **Wrong path, right endpoint.** Under live traffic the agent must yield, wait, or steer around vehicles absent from the demonstration. Such reactive detours are valid members of $\Phi(C)$ yet are penalized by ADE, Fréchet Distance, and nDTW. Conversely, a route that reaches a nearby pose without executing a required turn can score well on FDE while failing the command.

Per-metric verdicts. ADE is not reparameterization-invariant: it requires an index correspondence after resampling, so two agents driving the same path at different speeds receive different scores, even though the command never specifies speed. FDE checks only the terminal point and is silent on path constraints and legality (crossing a footpath or a red light can still yield zero FDE). Classical Fréchet Distance has the right invariance and detects worst-case excursions, but it is unbounded and sup-driven: aggregate means are set by a minority of catastrophic episodes, and mean versus median rankings can diverge (Section 4.6.3). ADE-Match, as implemented, is optimistically biased by the uniform-in- t degeneracy noted above. Pointing

Game and Recall@ k , used in earlier drafts for open-loop mask quality, measure a signal the closed-loop controller mostly discards; only the centroid of the largest connected component reaches the planner.

Closed-loop correspondence and noise floors. After the first divergence from τ^* , the agent occupies states the demonstration never visited. Subsequent ADE terms compare $\hat{\tau}$ to a counterfactual that no longer applies; this is the same distribution-shift phenomenon that motivates iterative imitation learning [103], applied here to evaluation rather than training. A goal predicate evaluated on the final state remains well defined regardless of how far the agent strayed. Two unmeasured noise floors further limit absolute interpretation of displacement scores: (i) demonstration and rollout share the same rule-based local planner, so much of the measured displacement is click-placement variance within the navigable region rather than model error; (ii) live traffic differs from the recorded realization, adding an irreducible stochastic term that no better model can remove.

What Task Completion costs. TC survives because success under a referential command is a predicate, and only a predicate can express membership in $\Phi(C)$. It is bounded, approximately binomial (hence Wilson intervals in Table 4.2), and comparable to outdoor-VLN literature [4, 98, 3]. Fréchet Distance and nDTW are retained as diagnostics rather than verdicts: both have reparameterization invariance, and nDTW is normalized to $[0, 1]$. The honest caveat is that human TC, as used in the ICRA protocol, *delegates* $\Phi(C)$ to a judge without recorded inter-rater agreement (Section 4.5). The construct problem was relocated into the annotator, not solved. Section 4.10.4 specifies a reproducible geometric success predicate that restores objectivity while still respecting multiple valid paths.

4.6 Experimental Results

We now report live-navigation results under the metrics above, starting with the CLIP family and then documenting the earlier architectural lineage.

4.6.1 Implementation Details

We use a CLIP ViT-B/32 backbone [56] pretrained on large-scale image-text pairs for feature extraction. Whether the CLIP encoders were frozen or fine-tuned during CARLA-NAV training was not documented in the archival paper or earlier drafts; we therefore do not claim a freeze/fine-tune protocol here. Fusion, trajectory encoding, and segmentation heads are trained on CARLA-NAV. When we say “trained from scratch” below, we refer specifically to the non-CLIP RNR variants (RNR-S / RNR-SC), whose visual backbones and task heads are randomly

initialized and trained only on CARLA-NAV, not that CLIP’s pretraining is discarded for CLIP-* models. Frames are selected with stride 10 and resized to 224×224 . Per-frame visual features have spatial resolution $H = W = 7$ and channel dimension $C_v = 512$. The historical contextual trajectory is plotted from episode start to the current timestamp, resized to 640×480 , and encoded to match the visual feature resolution. Linguistic features use the CLIP tokenizer and language encoder with maximum command length $N = 20$ and $C_l = 512$. We train with batch size 32 and AdamW [104], initial learning rate 10^{-4} , and polynomial learning-rate decay with power 0.5. For combo loss, $\lambda = 0.3$.

4.6.2 Baselines and Ablations

We compare our proposed approach CLIP-MC (multiple frames with contextual trajectory) against:

- RNR-S / RNR-SC: the RNR approach of Rufus et al. [5] with and without contextual trajectory, trained from scratch on CARLA-NAV;
- CLIP-S / CLIP-SC: the same RNR-style architecture with a CLIP backbone in place of the original visual encoder, without and with contextual trajectory. RNR-S vs. CLIP-S is therefore a clean backbone-only ablation under otherwise comparable single-frame grounding;
- CLIP-M: multiple frames without contextual trajectory;
- CLIP-MC: multiple frames with contextual trajectory (proposed).

As reported in Table 4.1, CLIP-MC outperforms all variants on Task Completion. Validation and test contain only 25 and 34 episodes respectively, so each validation episode corresponds to four percentage points of TC; reported deltas should be read with that granularity in mind. Introducing historical contextual trajectory improves validation TC by one episode for single-frame approaches (CLIP-SC 0.52 vs. CLIP-S 0.48) and by four episodes for multi-frame approaches (CLIP-MC 0.72 vs. CLIP-M 0.56). The multi-frame approach CLIP-M improves by two episodes over CLIP-S on both validation and test (0.56 / 0.55 vs. 0.48 / 0.47). These trends indicate that multiple frames and contextual trajectory are both helpful for outdoor VLN, but the absolute margins are modest relative to the evaluation size. CLIP-based single-frame models outperform non-CLIP RNR variants on the *test* split; we qualify the backbone comparison further when discussing path metrics below.

As shown in Table 4.3, CLIP-MC again outperforms all other variants by clear margins on Fréchet Distance (lower is better) and nDTW (higher is better). The remaining rows are more nuanced: on *validation*, RNR-SC (Fréchet 21.64, nDTW 0.45) outperforms CLIP-S, CLIP-SC, and CLIP-M on both path metrics, even though CLIP variants are stronger on test TC. We

Method	Task Completion	
	Val	Test
RNR-S	0.44	0.29
RNR-SC	0.52	0.32
CLIP-S	0.48	0.47
CLIP-SC	0.52	0.50
CLIP-M	0.56	0.55
CLIP-MC	0.72	0.68

Table 4.1: Task Completion (human-evaluated), demonstrating live navigation (Contribution 3). Validation has 25 episodes and test 34; CLIP-MC’s gains should be interpreted at that resolution (Section 4.10).

therefore do not claim a uniform backbone advantage from CLIP features alone; the benefit of CLIP is clearest when combined with multi-frame and trajectory context (CLIP-MC), and path-metric rankings can diverge from Task Completion on the small validation set, consistent with the outlier sensitivity of Fréchet Distance and nDTW noted in Section 4.5.

Moreover, preliminary evaluations of earlier models (Section 4.6.3) show that Fréchet Distance often has standard deviation *larger than the mean*, with the median roughly half the mean. Gaps of a few units among the CLIP variants in Table 4.3 therefore sit well inside typical episode-level dispersion; we do not treat those intermediate rankings as statistically established.

Table 4.4 studies the effect of the number of video frames for multi-frame models on Task Completion (Contribution 4). On the test split the trend is monotonic for both CLIP-MC and CLIP-M: TC rises steadily from $n = 1$ to $n = 8$. On validation the curve is non-monotonic; for both models $n = 2$ is one episode worse than $n = 1$ (CLIP-MC 0.48 vs. 0.52; CLIP-M 0.44 vs. 0.48) before recovering at larger n . Given the 25-episode validation set, a one-episode dip is within sampling noise and should not be over-interpreted. The best result remains $n = 8$ for CLIP-MC on both splits; we stop at eight frames for computational cost reasons discussed in Section 4.3.

4.6.3 Preliminary Architectures: ViT and Conv3D

Before the CLIP-MC family, we explored several earlier architectures. Chronologically the lineage is: GTA-V control regression $\rightarrow$ Cross-modal ViT (two-class mask) $\rightarrow$ Conv3D/X3D

Method	Val TC (Wilson 95% CI)		Test TC (Wilson 95% CI)	
	Estimate	CI	Estimate	CI
RNR-S	0.44	[0.27, 0.63]	0.29	[0.17, 0.46]
RNR-SC	0.52	[0.33, 0.70]	0.32	[0.19, 0.49]
CLIP-S	0.48	[0.30, 0.67]	0.47	[0.31, 0.63]
CLIP-SC	0.52	[0.33, 0.70]	0.50	[0.34, 0.66]
CLIP-M	0.56	[0.37, 0.73]	0.55	[0.39, 0.71]
CLIP-MC	0.72	[0.52, 0.86]	0.68	[0.51, 0.81]

Table 4.2: Wilson score 95% confidence intervals for Task Completion, derived from binomial counts on 25 validation and 34 test episodes. The intervals make the statistical fragility of single-run deltas explicit: CLIP-MC and weaker baselines overlap substantially on validation. Because TC is human-labeled, these intervals capture sampling uncertainty of the success counts but not annotator disagreement.

with joint self-attention $\rightarrow$ CLIP-MC. The ViT and Conv3D models share the multi-task mask formulation but use separate visual and language encoders. They were evaluated on an earlier 50-episode validation pool with displacement metrics (Section 4.5.1), not with the 25/34 TC protocol; we therefore report them as a separate preliminary evaluation and do *not* merge them into Tables 4.1–4.4.

4.6.3.1 GTA-V Control Regression

The earliest modeling attempt predicted continuous steering, acceleration, and brake from GTA V imagery (Chapter 3). That formulation was abandoned because low-level controls without an explicit spatial goal were brittle and because GTA V lacked the calibration access needed for dense navigable-region annotation. The move to CARLA and to mask prediction is the starting point of the lineage below.

4.6.3.2 Cross-Modal ViT (Two-Class Mask)

The first CARLA architecture used an ImageNet-pretrained Vision Transformer [45] (patch size 16×16 , hidden dimension $C = 384$, spatial features $H = W = 14$) with 300-dimensional GloVe embeddings [105] passed through an LSTM (maximum command length 20). Fusion was

Method	Fréchet Distance ↓		nDTW ↑	
	Val	Test	Val	Test
RNR-S	28.14	42.45	0.35	0.16
RNR-SC	21.64	44.65	0.45	0.33
CLIP-S	40.30	42.53	0.23	0.24
CLIP-SC	35.58	38.49	0.36	0.39
CLIP-M	32.92	53.10	0.39	0.26
CLIP-MC	13.54	15.06	0.54	0.59

Table 4.3: Fréchet Distance and nDTW (↓ lower is better; ↑ higher is better). CLIP-MC dominates both splits, but intermediate rankings among other rows should be read cautiously: path-metric dispersion often exceeds the mean (Section 4.6.3), so gaps of a few units sit inside episode-level noise.

pure cross-modal attention: with visual tokens $f_t \in \mathbb{R}^{HW \times C}$ and linguistic features $h \in \mathbb{R}^{N \times C}$,

$$\begin{aligned}
A &= \sigma(f_t \odot h^\top), \\
M &= A \odot h^\top, \\
f'_t &= M * f_t,
\end{aligned} \tag{4.5}$$

where σ is softmax, $\odot$ is matrix multiplication, and $*$ is element-wise modulation. Critically, the navigable-region head predicted *two* classes, intermediate region versus final goal, so that stopping could be decided from the final-class confidence. Live targeting used the largest connected component of the final class when confident, and otherwise the farthest point of the skeletonized trajectory mask as an intermediate waypoint; three consecutive final-class predictions ended the run. No quantitative results were published for this draft. It is the architectural ancestor of the ViT baseline below, which retained the per-frame ViT+LSTM stack but moved toward single-mask prediction with an area-based stop.

4.6.3.3 ViT Baseline (Per-Frame)

The evaluated baseline (Figure 4.2) keeps the ImageNet ViT and GloVe+LSTM encoders above. Visual and linguistic features are fused by multimodal attention; two ASPP-style heads predict the navigable-region and trajectory masks. Relative to the IROS draft, this variant collapses the two-class navigable head into a single mask and uses the area-threshold stopping rule of Algorithm 1 rather than final-class confidence.

Method	Split	Task Completion				
		$n = 1$	$n = 2$	$n = 4$	$n = 6$	$n = 8$
CLIP-MC	val	0.52	0.48	0.52	0.68	0.72
	test	0.50	0.53	0.56	0.62	0.68
CLIP-M	val	0.48	0.44	0.48	0.52	0.56
	test	0.47	0.47	0.50	0.53	0.55

Table 4.4: Ablation on the number of frames for multi-frame models on Task Completion (Contribution 4). On the test split the trend is monotonic from $n = 1$ to $n = 8$; the validation dip at $n = 2$ is one episode and should not be over-interpreted.

4.6.3.4 Conv3D (X3D) Model (Multi-Frame)

The multi-frame predecessor (Figure 4.3) uses an X3D backbone [106] pretrained on Kinetics-400 [107]. From $T = 40$ input frames it samples $T = 4$ frames at stride 10, extracts $C \times T \times H \times W$ features, and applies multimodal self-attention over concatenated visual and linguistic tokens (still GloVe+LSTM). A temporal convolution collapses the time axis before dual ASPP heads. Relative to the IROS draft, Conv3D upgrades fusion from pure cross-modal attention to *joint* self-attention over the multimodal sequence, and introduces multi-frame visual context.

4.6.3.5 Why CLIP

Three considerations motivated the migration from the ViT/Conv3D lineage to CLIP. First, the GloVe+LSTM language stack has no visual grounding: word vectors are trained on text alone, so alignment with image regions must be learned entirely inside the fusion block on CARLA-NAV. CLIP [56] instead provides a jointly pretrained image–text embedding space in which matched pairs already lie close together, giving the fusion module a stronger starting point for grounding (Section 2.5.4). Second, CARLA-NAV is small, on the order of 500 training episodes, so freezing (or lightly adapting) a large pretrained multimodal encoder is a better inductive bias than training fusion from scratch on unimodal GloVe features. Third, CLIP collapses the architecture: one dual encoder replaces the separate X3D/ViT visual backbone and LSTM language stack, simplifying training and ensuring that both modalities live in the same feature dimension ($C_v = C_l = 512$). Empirically, CLIP-MC is the published outcome of that migration (Tables 4.1–4.4). The ViT/Conv3D numbers below document the prior art within this project; they use a different evaluation pool and metric suite, so we make *no* cross-family numeric comparison.

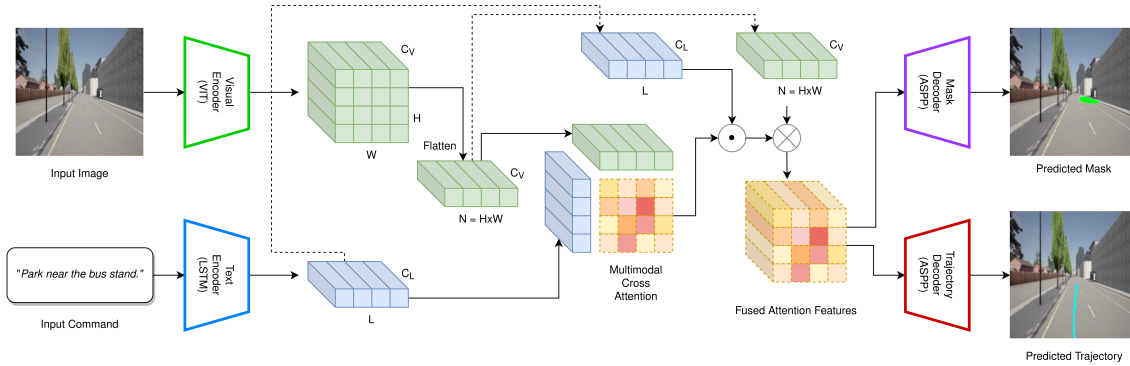


Figure 4.2: Per-frame ViT baseline that the CLIP migration replaced: ImageNet Vision Transformer, GloVe+LSTM language stack, multimodal attention, and dual ASPP heads. The encoders are pretrained separately, so alignment must be learned entirely inside the fusion block on CARLA-NAV.

Table 4.5 compares Conv3D and ViT on Fréchet Distance, FDE, ADE, and ADE-Match (mean / median / STD; meters). Conv3D is consistently better. Critically, the standard deviation often exceeds the mean (e.g., Conv3D Fréchet mean 17.84 with STD 21.49; median 9.10), and ViT is even more heavy-tailed (mean 43.95, STD 60.56, median 12.19). Path-metric means are therefore dominated by a minority of catastrophic episodes. Two further observations support the construct-validity critique in Section 4.5.2: (i) Conv3D Fréchet Distance and FDE nearly coincide (means 17.84 vs. 17.83; identical medians 9.10), so the Fréchet supremum is attained at the endpoint; the modal failure is right-path/wrong-endpoint, which ADE (10.32) dilutes and ADE-Match (4.73) hides; (ii) over the same episodes, FDE median 9.10 m and ADE-Match median 1.26 m cannot both be faithful summaries of median quality, underscoring internal inconsistency of the displacement suite.

4.6.3.6 Stratified Ablations

Tables 4.6–4.9 break the same evaluation down by command type, primary action verb, referred target type, and subcommand count (meters). Across strata, Conv3D remains stronger and more stable than ViT. Two qualitative patterns emerge: (i) stopping-based commands are harder than turning-based ones for ViT (Fréchet mean 53.98 vs. 36.42), while Conv3D is comparatively flat across both; (ii) two-subcommand episodes yield lower Conv3D errors than single-subcommand ones (ADE 6.43 vs. 12.13), consistent with intermediate clicks providing denser supervision. These stratified tables satisfy, for the ViT/Conv3D lineage, the per-maneuver breakdown that remains open for the CLIP family (Section 4.10). Cross-stratum comparisons should nevertheless be read cautiously: success predicates differ by command class,

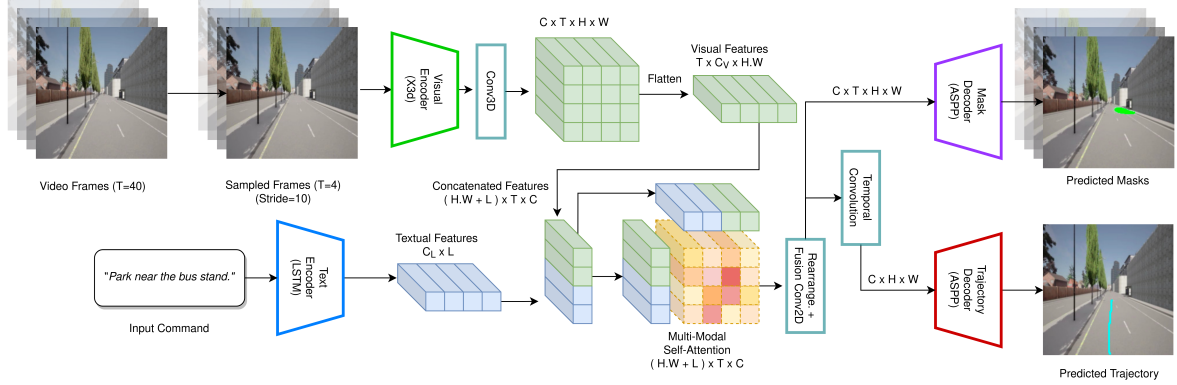


Figure 4.3: Multi-frame Conv3D (X3D) predecessor: sampled video frames through X3D, joint multimodal self-attention with LSTM language features, temporal convolution, and dual ASPP heads. Like the ViT baseline, it lacks a jointly pretrained vision–language embedding space.

Metric	Conv3D			ViT		
	Mean	Median	STD	Mean	Median	STD
Fréchet	17.84	9.10	21.49	43.95	12.19	60.56
FDE	17.83	9.10	21.50	39.88	12.19	51.38
ADE	10.32	6.61	11.04	22.50	7.50	34.00
ADE-Match	4.73	1.26	8.51	16.43	2.50	31.23

Table 4.5: Preliminary live-navigation results for Conv3D (X3D) vs. ViT on the earlier evaluation pool. All values are CARLA world-frame meters. ADE values follow the regenerated evaluation tables. Lower is better for all metrics.

endpoint-based for park/stop, state-based for lane change, sequence-based for multi-step, so a single displacement metric does not denote the same notion of difficulty in every row, and Fréchet $\approx$ FDE again holds in the turning stratum (both 17.97m for Conv3D).

4.6.4 Qualitative Results

Figure 4.4 compares CLIP-MC with RNR-S during live inference. Paths are overlaid on the aerial map of the CARLA environment. With multi-frame and historical trajectory context, CLIP-MC successfully performs turning and stopping maneuvers where the static-image RNR baseline fails. For “change to the left lane,” RNR-S continues straight while CLIP-MC changes

Type	Metric	Conv3D			ViT		
		Mean	Med.	STD	Mean	Med.	STD
Stopping	ADE	10.67	7.50	11.98	27.48	7.57	40.54
	ADE-Match	4.44	1.37	9.00	20.90	3.70	37.59
	FDE	17.32	9.67	21.38	47.67	12.28	60.74
	Fréchet	17.33	9.67	21.38	53.98	12.28	72.80
Turning	ADE	10.18	5.93	9.33	17.06	8.97	21.02
	ADE-Match	5.03	1.32	7.57	11.40	3.45	18.25
	FDE	17.97	6.97	21.37	35.57	16.68	39.78
	Fréchet	17.97	6.97	21.37	36.42	16.68	41.00

Table 4.6: Conv3D vs. ViT stratified by command type (stopping-based vs. turning-based). All displacement values are CARLA world-frame meters.

lane (with a slight delay). In a curved-road case near a traffic light, both methods stop early, mistaking the curve for an intersection, an instructive failure mode.

4.7 Failure Modes

Numbers alone do not reveal how the agent fails; the cases below catalog recurring live-episode patterns. Qualitative inspection of live episodes, including the cases shown in Figure 4.4, reveals recurring failure patterns. Figure 4.5 contrasts a successful live trajectory with a failure in which the predicted path overshoots the turn and ends far from the demonstration.

1. **Delayed lane change:** On commands such as “change to the left lane,” CLIP-MC eventually executes the maneuver but with a slight delay relative to the ground-truth path, while RNR-S may continue straight. Temporal context helps, but reaction lag remains.
2. **Curve mistaken for an intersection:** On a curved road approaching a traffic light, both CLIP-MC and RNR-S stop prematurely, treating the geometry as an intersection. This suggests sensitivity to road layout cues that resemble junction structure.
3. **Early stopping from the area threshold:** When several candidate goal regions are visible, the area-based stopping rule can fire before the intended final region is reached, because a large connected component satisfies the empirical threshold for five consecutive predictions. Area growth under perspective is a proxy for proximity, not for identity of the correct region; Section 4.8 returns to this point.

Action	Metric	Conv3D			ViT		
		Mean	Med.	STD	Mean	Med.	STD
stop	ADE	11.52	6.83	13.58	28.15	8.00	42.68
	Fréchet	14.90	10.87	16.82	55.34	16.65	74.14
go	ADE	8.63	4.90	9.27	31.52	4.84	54.59
	Fréchet	13.72	8.67	16.60	52.74	11.07	86.86
take	ADE	8.99	4.60	9.22	19.64	6.36	23.42
	Fréchet	12.81	5.77	18.17	45.52	9.60	54.31
turn	ADE	9.17	4.75	9.23	17.94	8.03	22.67
	Fréchet	15.79	6.13	20.79	35.70	11.28	42.20
change	ADE	9.23	4.10	11.11	10.67	12.16	5.49
	Fréchet	21.45	9.67	26.08	36.76	33.61	29.80
drive	ADE	10.18	4.90	10.70	19.48	4.91	32.91
	Fréchet	15.66	9.92	19.57	27.60	9.86	42.71
park	ADE	6.21	5.32	5.82	14.43	4.56	18.97
	Fréchet	18.49	7.03	28.67	41.97	8.42	73.67

Table 4.7: Conv3D vs. ViT stratified by primary action verb (ADE and Fréchet; meters). Full FDE/ADE-Match tables follow the same ranking and are omitted for space.

4. **Referential ambiguity and path metrics:** Commands such as “park near the blue dustbin” admit multiple valid stopping locations. Fréchet Distance and nDTW then penalize a spatially plausible path that differs from the recorded demonstration, even when Task Completion may still succeed.
5. **Synthetic-to-real domain gap:** All training and evaluation are in CARLA. Appearance, sensor noise, and traffic behavior in the real world differ, so transfer without domain adaptation [108, 109] is an open risk (Section 4.9).

4.8 Discussion

The tables above establish what happens; this section asks why, why it is interesting, and why it is useful. Failure patterns themselves are catalogued in the previous section and are not repeated here.

Target	Metric	Conv3D			ViT		
		Mean	Med.	STD	Mean	Med.	STD
infrastructure	ADE	11.62	7.29	14.30	32.78	6.54	46.15
	ADE-Match	4.84	1.13	10.66	26.85	3.02	44.47
	FDE	16.79	9.44	19.27	55.21	12.04	70.36
	Fréchet	16.81	9.44	19.26	64.85	12.04	85.24
road/light	ADE	8.86	4.60	9.15	20.13	6.36	32.04
	ADE-Match	4.21	1.07	6.94	14.29	1.72	29.59
	FDE	15.64	7.52	19.48	33.31	12.69	37.97
	Fréchet	15.64	7.52	19.48	38.85	12.69	53.44
automobile	ADE	10.58	8.32	4.97	19.24	8.51	21.30
	ADE-Match	4.15	2.16	5.29	14.06	4.61	20.67
	FDE	25.48	13.31	34.69	47.48	22.17	45.58
	Fréchet	25.48	13.31	34.69	47.48	22.17	45.58

Table 4.8: Conv3D vs. ViT stratified by referred target type. All displacement values are CARLA world-frame meters.

Trajectory context helps most with multiple frames. Historical contextual trajectory improves validation TC by one episode for single-frame models (CLIP-SC 0.52 vs. CLIP-S 0.48) but by four episodes once multi-frame visual input is present (CLIP-MC 0.72 vs. CLIP-M 0.56). Why? A single frame has no way to distinguish “before the turn” from “after the turn,” so the past-trajectory map is the only progress signal available; with eight frames the model can also read motion, and the two turn out to be complementary rather than redundant. This is interesting because it tracks instruction progress without an explicit instruction-segmentation module. It is useful because it suggests that progress context and short-term visual history should be designed as a pair, not as interchangeable substitutes.

Fréchet Distance $\approx$ FDE is a diagnosis, not a coincidence. For Conv3D, Fréchet Distance and FDE means nearly coincide (17.84 vs. 17.83 m) with identical medians (9.10). The supremum is attained at the terminal point in essentially every episode, so the modal failure is right-path/wrong-endpoint. That is interesting because it tells a follow-up to spend effort on the stopping rule rather than on path following. It is also the direct empirical justification for the geometric goal-region predicate in Section 4.10.4: if the path is usually fine and the endpoint is the problem, a predicate on the final pose is the right primary metric.

# Subs	Metric	Conv3D			ViT		
		Mean	Med.	STD	Mean	Med.	STD
1	ADE	12.13	7.22	12.93	23.26	8.00	30.82
	ADE-Match	6.05	1.29	9.95	17.30	3.82	28.93
	FDE	21.69	10.58	24.58	44.78	13.15	55.70
	Fréchet	21.70	10.58	24.57	46.32	13.15	58.77
2	ADE	6.43	4.66	5.13	19.64	4.74	38.49
	ADE-Match	1.91	1.19	2.49	14.60	1.26	36.61
	FDE	9.64	7.50	8.65	29.45	9.11	40.34
	Fréchet	9.64	7.50	8.65	38.91	9.11	65.90

Table 4.9: Conv3D vs. ViT stratified by number of subcommands. All displacement values are CARLA world-frame meters.

Why RNR-SC can win on validation path metrics yet lose on test TC. On validation, RNR-SC outperforms CLIP-S, CLIP-SC, and CLIP-M on Fréchet Distance and nDTW, even though CLIP variants are stronger on test Task Completion. Two competing hypotheses remain open. First, sup-driven metrics on 25 episodes can be dominated by one catastrophic outlier, so a backbone that avoids a single bad episode looks stronger without being so on average. Second, CLIP may ground better across maps (hence higher test TC) while driving less smoothly on the small validation pool (hence worse path metrics). The first hypothesis is testable by reporting medians and bootstrap CIs; the second requires a map-unseen protocol (Section 4.10). We do not assert either; the discrepancy is itself a reason not to over-interpret intermediate path-metric rankings.

Two-subcommand episodes score better than single-subcommand ones. Conv3D ADE is 6.43m on two-subcommand episodes versus 12.13m on single-subcommand ones. The denser-supervision explanation (intermediate clicks supply more mask targets) is plausible, but it is confounded with episode length and command difficulty. A length-controlled check would be needed before treating this as a design recommendation.

Path-metric dispersion sets a noise floor on Table 4.3. Standard deviations often exceed the means in Table 4.5. Gaps of a few units among non-CLIP-MC rows in Table 4.3 therefore sit inside episode-level dispersion. Aggregate path metrics remain useful for spotting the CLIP-MC jump; they are a poor basis for ranking the intermediate rows.

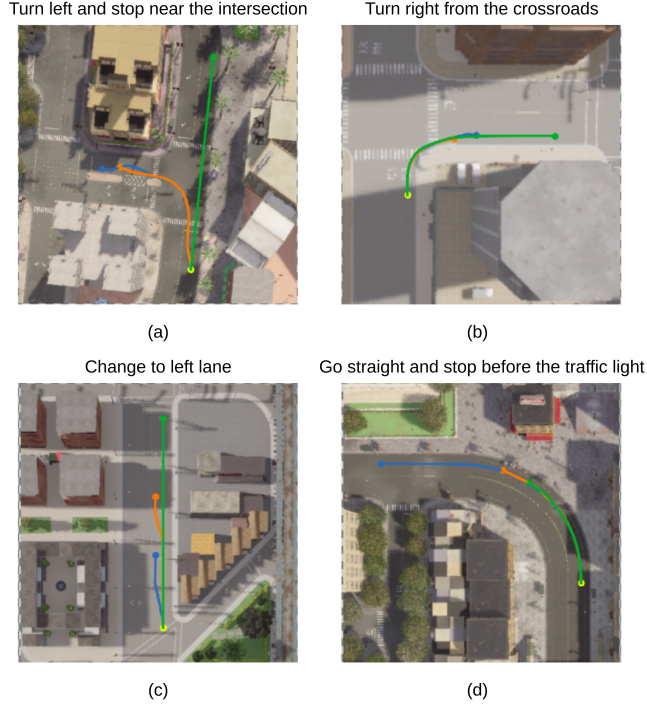


Figure 4.4: Qualitative live navigation on CARLA-NAV (yellow = start, orange = CLIP-MC, green = RNR-S, blue = ground truth). The lane-change case shows temporal context changing behavior; the curved-road case near a traffic light shows where both methods still fail by mistaking the curve for an intersection.

Area-threshold stopping fires on size, not identity. As noted under Failure Modes, the area rule can stop early when several candidate regions are visible. Area growth under perspective proxies proximity, not identity of the referred region. Future stopping rules should therefore combine area with semantic confirmation of referred objects or confidence-aware temporal consistency [98].

Set-valued targets unify the metric and the head. The same observation that a command designates a *set* $\Phi(C)$ of valid trajectories (Section 4.5.2) explains both why Task Completion is the primary metric and why the output head is a segmentation map rather than a coordinate regressor. A regressor trained under squared loss converges to the conditional mean of the valid modes; for “park between the two cars,” that mean sits in the middle of the road and is not itself a valid goal. A segmentation head places mass on multiple disjoint valid regions and recovers a mode via the largest connected component. The thesis previously applied this insight only to evaluation; stating it here connects the metric argument to the architectural one.

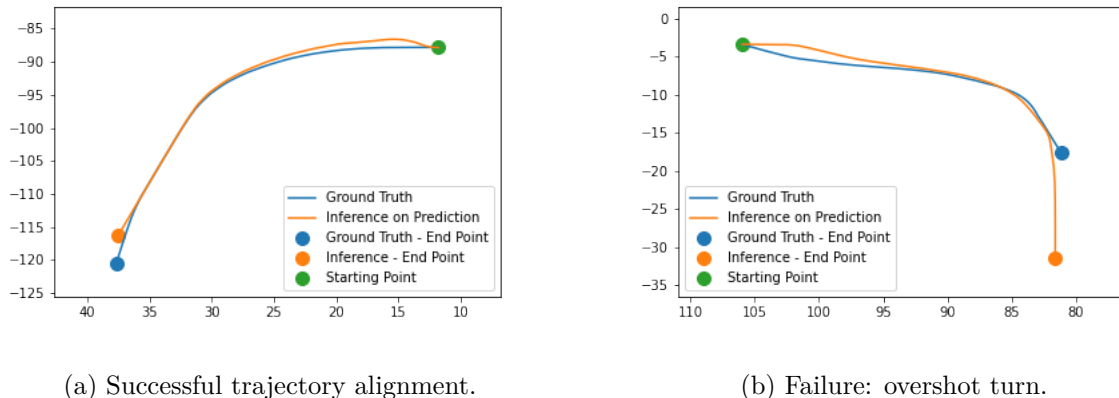


Figure 4.5: Bird’s-eye comparison of ground-truth (blue) and live inferred (orange) trajectories (green = start; colored dots mark endpoints). The failure is right-path/wrong-endpoint: the same pattern that Fréchet $\approx$ FDE quantifies in Table 4.5 and that motivates Task Completion as the primary metric.

4.9 Limitations

The discussion above already flags stopping-rule and metric issues; three broader limitations constrain real-world claims. A major limitation that restricts real-world practicality is the synthetic nature of CARLA-NAV; domain adaptation [108, 109] is required before claiming transfer to real imagery. The stopping criterion is rule-based rather than learned; learning when to stop [98] is more scalable (see Failure Modes and Discussion). Finally, the predicted future trajectory currently serves interpretability only; incorporating it into the planner for end-to-end navigation remains future work. These limitations were drafted for an earlier conference submission and omitted from the archival six-page version for space; we restore them here for thesis-level completeness.

4.10 Extended Evaluation Protocol

The ICRA evaluation protocol underlying Tables 4.1–4.4 is limited in several ways that matter for a thesis-level claim. We document those limits, report what can be derived without re-training (Wilson CIs above; ViT/Conv3D stratified tables), and specify the protocols we adopt for stronger evaluation. Where new live runs were not available for this manuscript, we state the intended measurement explicitly rather than implying the current tables already cover it.

4.10.1 Held-Out Map Split

The default CARLA-NAV splits shuffle episodes across eight maps, so train, validation, and test can share map layouts (Chapter 3). Following the seen/unseen convention of R2R [1], we define a **map-unseen** protocol: train on six maps, validate on one held-out map, and test on the remaining map (rotated so each map appears in test across folds). Under this protocol, Task Completion, Fréchet Distance, and nDTW should be reported separately for map-seen and map-unseen episodes. The numbers in Tables 4.1–4.4 use the original shuffled split and therefore do *not* establish cross-map generalization; they are an upper bound relative to a true unseen-map evaluation.

4.10.2 Multi-Seed Confidence and Stochastic Live Traffic

Live CARLA episodes are stochastic under dynamic traffic. A single evaluation pass yields the point estimates above; Table 4.2 adds Wilson intervals for that pass, but does not capture seed-to-seed variance of training or of traffic randomization, nor human-annotator disagreement on TC (see Section 4.5 for the same caveat). The recommended protocol is: train with at least three random seeds, evaluate each checkpoint on the live val/test suites with fixed spawn poses but resampled traffic, and report mean $\pm$ standard deviation of TC together with bootstrap CIs for path metrics. Overlapping Wilson intervals for CLIP-MC vs. CLIP-M on validation already caution against overstating single-run deltas.

4.10.3 Safety and Infractions

Task Completion records goal arrival but not *how* the vehicle drove. We therefore include CARLA’s built-in sensors in the evaluation contract:

- **Collisions:** count of collision events (vehicles, pedestrians, static props) per episode and per kilometer.
- **Lane invasions:** cumulative lane-invasion sensor triggers.
- **Traffic-light infractions:** red-light violations recorded by the simulator’s traffic-light API.

An episode may count as Task Completion success yet still be unacceptable under these rates. Infractions were not logged in the original ICRA runs; subsequent live evaluation should report them alongside TC.

4.10.4 Reproducible Goal-Region Success Predicate

Human TC relocates the construct problem into the annotator (Section 4.5.2). CARLA-NAV already stores annotated final and intermediate navigable regions, so success can be redefined as a reproducible geometric predicate that still admits multiple valid paths:

1. **Goal arrival:** the final vehicle pose lies within radius r of the annotated final navigable region (equivalently, within the region or its r -dilation in world coordinates).
2. **Subcommand constraints:** for multi-step commands, the trajectory visits each annotated intermediate navigable region (again up to radius r) in order.
3. **Legality:** the episode incurs zero collisions and zero red-light infractions under the CARLA sensors listed above; lane invasions may be reported separately rather than hard-failed, depending on the maneuver class.

An episode succeeds only if all three hold. The radius r should be chosen on a held-out calibration set and reported; a natural default is on the order of the final-region diameter used during annotation. This predicate answers both halves of the right-path/wrong-endpoint dichotomy: the endpoint is checked against the annotated goal set rather than a single demonstration pose, and intermediate regions encode the path constraints that FDE alone misses.

Calibration experiments. Two measurements are needed before displacement scores or geometric TC can be interpreted absolutely: (i) an *annotation noise floor*, re-collect a subset of episodes with a second navigator and report ADE/FDE between the two human trajectories under matched traffic seeds, estimating click-placement variance; (ii) a *traffic stochasticity floor*, roll out a fixed checkpoint repeatedly from the same spawn with resampled traffic and report the dispersion of path metrics and geometric TC. Neither was measured in the ICRA runs; both belong in any follow-up that claims absolute navigation quality.

4.10.5 Trajectory-Head Ablation, Per-Maneuver Breakdown, and Runtime

- **Trajectory head:** Train a navigation-only variant (CLIP-MC without the z_t head) and compare TC and path metrics to the full multi-task model. Separately score the trajectory head with soft Dice / IoU against projected ground-truth short-horizon paths. Without this ablation, Contribution 2 rests on interpretability alone (as noted in Section 1.4).
- **Per-maneuver breakdown:** For the ViT/Conv3D lineage, Tables 4.6–4.9 already provide stratified displacement metrics by command type, action verb, target type, and subcommand count. The corresponding TC-stratified breakdown for the CLIP family (turn, straight, stop, park, multi-step) remains future work; aggregate CLIP metrics can still hide systematic failure on parking or lane-change commands.

- **Runtime:** Profile wall-clock milliseconds per forward pass (CLIP encode + fusion + heads) and end-to-end control frequency (Hz) on the live inference machine, with and without the every-5-frame schedule. The complexity argument in Section 4.3 predicts quadratic growth in T ; measured latency makes that claim concrete.

4.10.6 Ethics and Deployment Safety

Language-guided driving systems can fail silently under referential ambiguity, domain shift, or planner error. We do not claim readiness for real vehicles: all results are in simulation, collision metrics are not yet reported, and synthetic-to-real transfer remains open. Any deployment path would require redundant safety layers (collision checking, human override, and conservative stopping) independent of the grounding network.

4.11 Chapter Summary

We reformulated RNR for dynamic outdoor VLN, motivated the grounding-first design formally, introduced a multi-task network with CLIP features, multi-frame context, and historical trajectory, and demonstrated live navigation in CARLA. We also documented the preliminary ViT and Conv3D architectures and their stratified displacement results, and argued that Task Completion, not ADE/FDE/ADE-Match, is the appropriate primary metric because language commands specify sets of valid trajectories rather than a single demonstration path. Quantitative CLIP results show that CLIP-MC outperforms single-frame and non-contextual baselines on Task Completion, Fréchet Distance, and nDTW, with Wilson intervals, human-TC disclosure, path-metric dispersion caveats, and the extended evaluation protocol clarifying the statistical and safety limits of those gains. The next chapter concludes the thesis and outlines future work.

Chapter 5

Conclusion

This thesis addressed Vision-and-Language Navigation for autonomous driving in dynamic outdoor environments. We argued that existing approaches—graph-based indoor VLN, discrete-action outdoor VLN, and single-image navigable-region grounding—are insufficient when the referred goal is not yet visible, when fine maneuvers are required, and when navigation must be evaluated live in a changing scene.

We introduced CARLA-NAV (Contribution 1), a meta-dataset of language-guided outdoor navigation episodes in the CARLA simulator, with dense annotations for intermediate and final navigable regions and short-term future trajectories. The observer–navigator–verifier protocol, combined with automated projection of clicked regions into segmentation masks, enabled scalable curation across multiple maps and weather conditions.

We then argued formally for a grounding-first factorization, dense navigable-region prediction plus a fixed local planner, and proposed a multi-task network (Contribution 2) that predicts navigable-region and future-trajectory masks from multi-frame CLIP features, a historical contextual trajectory, and the language command. Predicted masks are used for live navigation (Contribution 3) with an empirical area-based stopping rule (five consecutive predictions above a threshold whose numeric value was not recorded). On the 25/34-episode validation/test splits, CLIP-MC reaches human-evaluated Task Completion of 0.72 / 0.68 (Wilson 95% CIs [0.52, 0.86] / [0.51, 0.81]), outperforming single-frame CLIP and RNR baselines on aggregate metrics, with corresponding improvements in Fréchet Distance and nDTW. Preliminary ViT and Conv3D (X3D) models, evaluated with Fréchet/FDE/ADE/ADE-Match and stratified by command type, action, target, and subcommand count, document the earlier architectural lineage and quantify path-metric dispersion (Contribution 4); we argue that Task Completion is the appropriate primary metric because commands specify sets of valid trajectories rather than a single demonstration path. We document the limits of these gains (sample size, human TC labeling, shared-map splits, missing safety metrics) and specify an extended evaluation protocol, including a reproducible goal-region success predicate, for follow-up.

Overall, grounding-first control provides an interpretable and continuous-control alternative to discrete VLN formulations, and is a promising direction for language-conditioned autonomous driving.

5.1 Ethics and Deployment Safety

This work studies language-conditioned control in simulation only. We do not claim that the system is safe for real vehicles: collision, lane-invasion, and traffic-light metrics are specified in Chapter 4 but were not part of the original evaluation; synthetic-to-real appearance and traffic gaps remain large; and referential commands can be ambiguous. Any path toward deployment would require independent safety monitors, human override, and evaluation under adversarial and long-tail scenarios beyond CARLA-NAV.

5.2 Future Work

- **Map-unseen and multi-seed evaluation:** Execute the held-out-map protocol and multi-seed live runs with Wilson/bootstrap intervals described in Chapter 4.
- **Safety metrics:** Log CARLA collisions, lane invasions, and red-light infractions alongside Task Completion.
- **Trajectory-head ablation and runtime:** Quantify whether the auxiliary trajectory head improves navigation, and profile inference latency versus frame count T .
- **CLIP per-maneuver TC:** Extend the ViT/Conv3D stratified tables to Task Completion for the CLIP family.
- **Real-world transfer:** Domain adaptation techniques [108, 109] are needed before deployment on real vehicles (Section 4.9).
- **Learnable stopping:** Replacing the rule-based area threshold with a learned stopping policy [98] should improve robustness (Section 4.8).
- **Trajectory-conditioned control:** The predicted future trajectory is currently used for interpretability; incorporating it into the planner could improve end-to-end behavior.
- **Richer evaluation:** Adopt the reproducible goal-region success predicate of Chapter 4 (final pose within radius r of the annotated final region, ordered intermediate-region visitation, and legality checks), and measure the annotation and traffic-stochasticity noise floors that currently bound interpretation of displacement metrics.

- **Interactive language:** Extending the setting to clarification dialogues and multi-turn instructions would bring the system closer to human co-driving.

Related Publications

1. Jain, K., Chhangani, V., Tiwari, A., Krishna, K. M., & Gandhi, V. (2023). Ground then Navigate: Language-guided Navigation in Dynamic Scenes. *IEEE International Conference on Robotics and Automation (ICRA) 2023*.

Author contribution statement. The ICRA 2023 paper above is the archival publication corresponding to this thesis. Kanishk Jain and Varun Chhangani are listed as equal first authors on that paper. For the work reported here, the candidate’s contributions include: (i) design and implementation of the CARLA-NAV data-collection toolkit and a substantial share of episode annotation in the observer, navigator, and verifier roles; (ii) development of the live-navigation pipeline that converts predicted navigable-region masks into planner targets and applies the area-based stopping criterion; (iii) training and evaluation of the CLIP-based multi-frame variants and ablations reported in Chapter 4; and (iv) analysis of failure modes and evaluation limitations. Co-authors contributed complementary modeling components, additional annotation, and advising. Dataset curation was performed jointly by the candidate, Amogh Tiwari, and Kanishk Jain (with KV Aditya Srivatsa assisting early tooling), with Dr. Vineet Gandhi providing complex commands and verification in ambiguous cases; this reconciles the collection-team description in Chapter 3 with the ICRA author list.

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